



US012404480B2

(12) **United States Patent**
Teffenhart et al.

(10) **Patent No.:** **US 12,404,480 B2**

(45) **Date of Patent:** **Sep. 2, 2025**

(54) **POLYUREA MICROCAPSULES AND LIQUID SURFACTANT SYSTEMS CONTAINING THEM**

C11D 3/37 (2006.01)

C11D 3/50 (2006.01)

(Continued)

(71) Applicant: **SYMRISE AG**, Holzminden (DE)

(52) **U.S. Cl.**

CPC *C11D 3/505* (2013.01); *B01J 13/16* (2013.01); *C11D 3/3726* (2013.01); *C11D 11/0094* (2013.01); *C11D 17/0013* (2013.01); *C11D 17/0039* (2013.01)

(72) Inventors: **John Teffenhart**, Edison, NJ (US);
Timothy Evans-Lora, Brooklyn, NY (US); **Keith McDermott**, Bound Brook, NJ (US)

(58) **Field of Classification Search**

CPC C11D 3/323; C11D 3/26; C11D 3/3726; C11D 3/50; C11D 3/505; C11D 11/0064; C11D 11/0094; C11D 17/0013; C11D 17/0039; B01J 13/14; B01J 13/16

(73) Assignee: **SYMRISE AG**, Holzminden (DE)

(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 223 days.

See application file for complete search history.

(56)

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Primary Examiner — Brian P Mruk

(74) *Attorney, Agent, or Firm* — Dilworth & Barrese, LLP

(57)

ABSTRACT

Suggested are polyurea microcapsules obtainable or obtained by reacting di- and/or polyisocyanates and guanidine carbonate in the presence of chemically modified biopolymer colloidal protection agents. The capsules can be loaded for example with fragrances and show high resistance against agglomeration in unstructured (viscosity 10-300 cps), anionic surfactant based liquid products.

13 Claims, 3 Drawing Sheets

Related U.S. Application Data

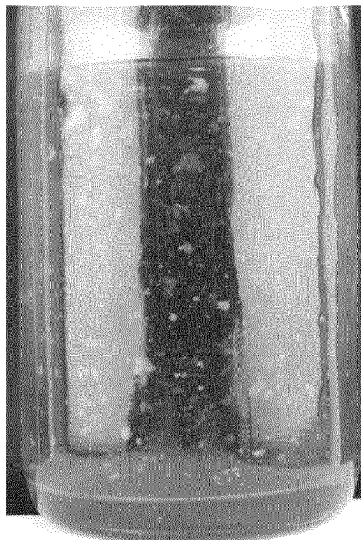
(60) Provisional application No. 62/915,811, filed on Oct. 16, 2019.

(51) **Int. Cl.**

C11D 3/26 (2006.01)

B01J 13/14 (2006.01)

B01J 13/16 (2006.01)



- (51) **Int. Cl.**
C11D 11/00 (2006.01)
C11D 17/00 (2006.01)

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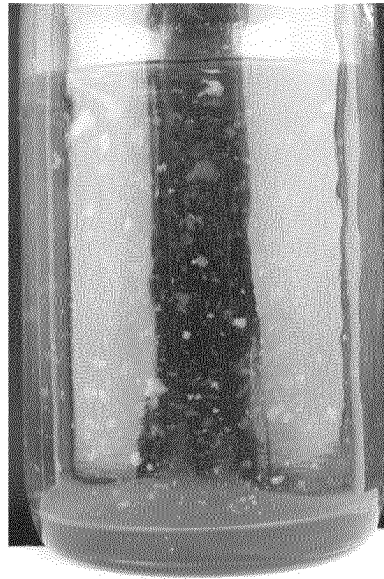


Figure 1



Figure 2

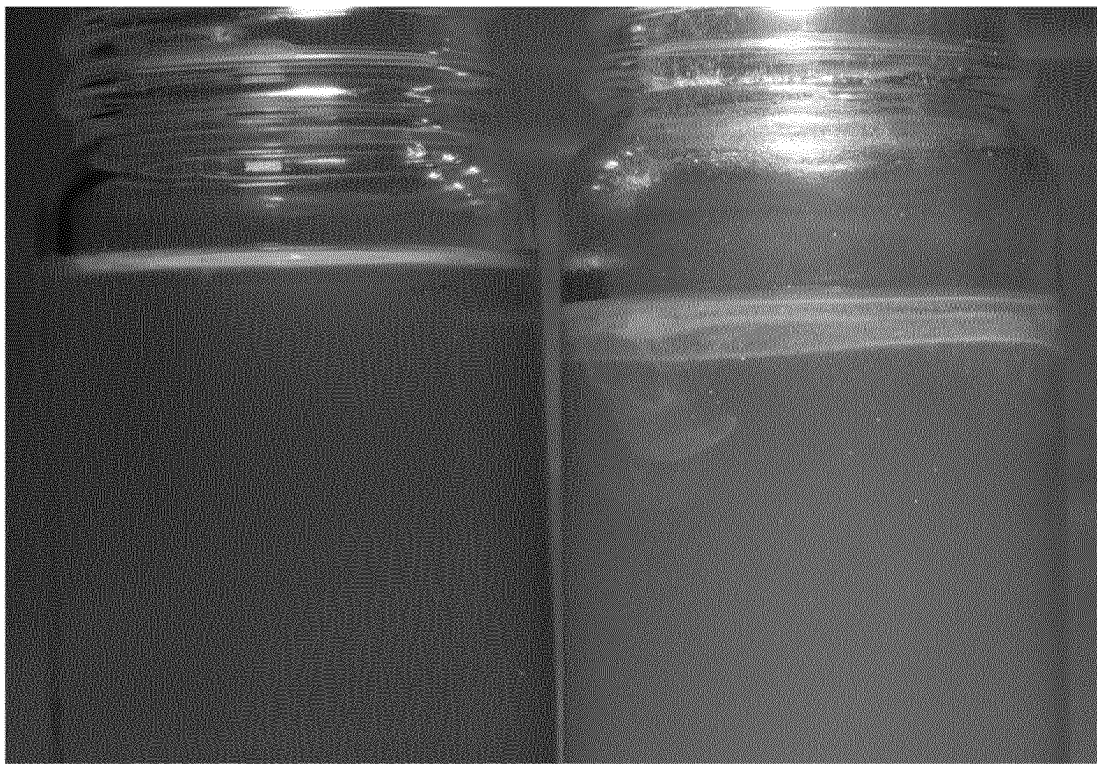


Figure 3



Figure 4

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POLYUREA MICROCAPSULES AND LIQUID SURFACTANT SYSTEMS CONTAINING THEM

This application is a 371 of PCT/EP2020/051124 filed Jan. 17, 2020, which claims benefit to Provisional Ser. No. 62/915,811, filed Oct. 16, 2019.

AREA OF INVENTION

The present invention refers to the area of microcapsules and covers microcapsules of polyurea-type with improved resistance against agglomeration and liquid surfactant systems containing them.

BACKGROUND OF THE INVENTION

There are many ways of producing micro encapsulated fragrance, each different process changes the properties of the capsules produced. Some applications are sensitive to capsules made with many colloidal protection agents. Applications such as unstructured liquid laundry detergents and body washes, suffer from capsule agglomeration overtime when most colloidal agents are used.

An example of this agglomeration is shown in FIG. 1 where Symcap® G type capsules are shown disbursed at 0.3% w/w in commercially available unstructured liquid laundry detergent. The sample is aged in an oven at 40° C. for 1 week to approximate 4 weeks storage at room temperature.

RELEVANT PRIOR ART

WO 2018 002214 A1 (FIRMENICH) discloses a method for making a core-composite shell microcapsule slurry for the delivery of hydrophobic active ingredients such as fragrance components of perfume oils. The method includes forming an outer shell by coacervation surrounding an internal phase which contains the hydrophobic active ingredient; and forming an inner shell by interfacial polymerization at the interface between the internal phase and the outer shell. The internal phase contains the hydrophobic active ingredient. The microcapsules are typically incorporated in a consumer product wherein the composite shell prevents the hydrophobic active ingredient from release until desired, generally during use of the consumer product.

WO 2018 019 894 A1 (FIRMENICH) relates to a process for the preparation of microcapsules comprising the steps of: 1) admixing a hydrophobic active ingredient with at least one polyisocyanate having at least three isocyanate functional groups to form an oil phase, provided that the oil phase is essentially free from diisocyanate; 2) dissolving an ionic or nonionic emulsifier in water to form a water phase, wherein the ionic emulsifier is chosen in the group consisting of gum Arabic, carboxymethyl cellulose, soy protein, sodium caseinate, gelatin, bovine serum albumin, sugar beet pectin, hydrolyzed soy protein, hydrolyzed sericin, Pseudo collagen, Biopolymer SA-N, Pentacare-NA PF, and mixtures thereof and wherein the non-ionic emulsifier is chosen in the group consisting of polyvinyl alcohol, modified polyvinyl alcohol, modified starch, modified cellulose, polysaccharides and mixtures thereof; 3) adding the oil phase to the water phase to form an oil-in-water dispersion; 4) applying conditions sufficient to induce interfacial polymerisation and form microcapsules in form of a slurry.

US 2013 0337023 A (IFF) claims a method for preparing a polyurea capsule composition comprising (a) preparing an

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oil phase comprising an active material and a polyisocyanate; (b) preparing a surfactant solution; (c) emulsifying the oil phase into the surfactant solution to form a fragrance emulsion; (d) adding a cross-linking agent to the fragrance emulsion to form a capsule slurry; and (e) curing the capsule slurry.

OBJECT OF THE INVENTION

Capsules with a polyurea shell have been found rather stable even in aqueous solutions with high concentrations of preferably anionic surfactants—as for example liquid detergents and softeners. However, polyurea microcapsules also show a serious tendency to form agglomerates particularly when introduced in aqueous anionic surface active compositions over time and temperature aged stability. Therefore, it has been the object of the present invention providing modified polyurea microcapsules with improved resistance against the formation of agglomerates, particularly in unstructured products that contain anionic surfactants (e.g. common liquid laundry preparations).

BRIEF DESCRIPTION OF THE INVENTION

A first object of the present invention refers to polyurea microcapsules obtainable or obtained according to the following steps:

- (a) providing an oil phase comprising at least one aliphatic and/or aromatic di- and/or polyisocyanate and optionally one or more actives for encapsulation and/or at least one oil component;
- (b) providing a first aqueous phase comprising at least one modified biopolymer colloidal protection agent and optionally at least one non-interfering emulsifier;
- (c) providing a second aqueous phase comprising guanidine carbonate and optionally at least one cationic surface active agent;
- (d) blending the oil phase and the first aqueous phase to form an emulsion;
- (e) adding the second aqueous phase to the emulsion formed in step (d) to form a dispersion of crude microcapsules (d);
- (f) curing the mixture; and optionally
- (g) removing the solvent(s) to obtain the microcapsules.

Another object of the present invention covers the corresponding for making polyurea microcapsules, comprising or consisting of the following steps:

- (a) providing an oil phase comprising at least one aliphatic and/or aromatic di- and/or polyisocyanate and optionally at least one oil component;
- (b) providing a first aqueous phase comprising at least one modified biopolymer colloidal protection agent;
- (c) providing a second aqueous phase comprising guanidine carbonate;
- (d) blending the oil phase and the first aqueous phase to form an emulsion;
- (e) adding the second aqueous phase to the emulsion formed in step (d) to form a dispersion of crude microcapsules;
- (f) curing the mixture; and optionally
- (g) removing the solvent(s) to obtain the microcapsules.

Surprisingly it has been found that incorporating specific modified biopolymer colloidal protection agents, particularly chemically modified biopolymers such as starches, gum Arabic and modified celluloses reduce tendency for agglomeration of polyurea capsules significantly. Di- and Polyisocyanates

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The formation of the polyurea capsule shell takes place by reaction of a compound having at least two isocyanate group and guanidine carbonate which has four amine groups. In a preferred manner the isocyanates contain two, three or more aliphatic and/or aromatic isocyanate groups. While the diisocyanate compound can be of aliphatic origin, the preferred embodiments encompass aromatic or cycloaliphatic compounds or their mixtures, such as for example

methylenediphenyl diisocyanate (MDI);
toluol diisocyanate (TDI);
hexamethylene diisocyanate (HDI);
isophorone diisocyanate (IPDI);
4,4-dicyclohexylmethan diisocyanate (H12MDI)

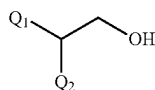
and mixtures thereof.

Particularly preferred is toluylene diisocyanate (also cited as toluol diisocyanate) which is available for example under the trademark Desmodur® (COVESTRO) in the market. With regard to the performance of the end product a toluylene diisocyanate encompassing about 50 to about 80 mol-% of the 2,4-isomer is particularly preferred.

Oil Components

The oil phase may consist of the isocyanate component itself, preferably however it contains one or more oil components as a solvent with C log P (Octanol:water Partition) values greater than 4, such as for example:

- (i) linear or branched saturated paraffins (mineral oils) having 15 or more C atoms, in particular having 18 to 45 C atoms;
- (ii) esters having 12 or more C atoms of linear or branched fatty acids having 6 to 30 C atoms and linear or branched, saturated or unsaturated mono-, di- or triols having 3 to 30 C atoms, these esters having no free hydroxyl groups;
- (iii) esters of benzoic acid and linear or branched, saturated or unsaturated monoalkanols having 8 to 20 C atoms;
- (iv) monoesters or diesters of alcohols having 3 to 30 C atoms and naphthalene-monocarboxylic or -dicarboxylic acids; especially naphthalenemonocarboxylic acid C₆-C₁₈ esters and naphthalenedicarboxylic acid di-C₆-C₁₈ esters;
- (v) linear or branched, saturated or unsaturated di-C₆-C₁₃-alkyl ethers;
- (vi) silicone oils;
- (vii) 2-alkyl-1-alkanols of the formula (III)



where

Q₁ is a linear or branched alkyl radical having 6 to 24 C atoms and

Q₂ is a linear or branched alkyl radical having 4 to 16 C atoms.

An oil phase or oil component in the narrower (and preferred) sense of the present invention, i.e. of the inventively limited substances or substances present only in a minor fraction, encompasses the following groups of substances:

- (i) linear or branched, saturated paraffins having 20 to 32 C atoms;
- (ii) esters having at least 14 C atoms of linear or branched, saturated fatty acids having 8 to 24 C atoms and linear

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or branched, saturated or unsaturated mono-, di- or triols having 3 to 24 C atoms, these esters containing no free hydroxyl groups;

- (iii) esters of benzoic acid and linear or branched, saturated monoalkanols having 10 to 18 C atoms;
- (iv) Alkylenediol dicaprylate caprates especially propylene-diol dicaprylate caprate;
- (v) linear or branched, saturated di-C₆-C₁₈-alkyl ethers, especially (straight-chain) di-C₆-C₁₂-alkyl ethers;
- (vi) silicone oils from the group of the cyclotrisiloxanes, cyclopentasiloxanes, dimethylpolysiloxanes, diethylpolysiloxanes, methylphenylpolysiloxanes, diphenylpolysiloxanes and hybrid forms thereof;
- (vii) 2-alkyl-1-alkanols having 12 to 32 C atoms of the formula (III) where Q₁ is a (preferably linear) alkyl radical having 6 to 18 C atoms and Q₂ is a (preferably linear) alkyl radical having 4 to 16 C atoms.

An oil phase in the narrowest (and most preferred) sense of the present invention encompasses the following groups of substances:

- (i) linear or branched, saturated paraffins having 20 to 32 C atoms such as isoeicosane or squalane;
- (ii) esters having at least 16 C atoms of linear or branched, saturated fatty acids having 8 to 18 C atoms and linear or branched, saturated mono-, di- or triols having 3 to 18 C atoms, these esters containing no free hydroxyl groups;
- (iii) esters of benzoic acid and linear or branched, saturated monoalkanols having 12 to 15 C atoms, especially C₁₂₋₁₅-alkyl benzoates;
- (iv) Alkylenediol dicaprylate caprates especially propylene-diol dicaprylate caprate;
- (v) straight-chain di-C₆-C₁₀-alkyl ethers; especially di-n-octyl ether (dicaprylyl ether);
- (vi) silicone oils from the group undecamethylcyclotrisiloxane, cyclomethicone, decamethylcyclopentasiloxane, dimethylpolysiloxanes, diethylpolysiloxanes, methylphenylpolysiloxanes and diphenylpolysiloxanes;
- (vii) 2-alkyl-1-alkanols having 12 to 32 C atoms of the formula (III) where Q₁ is a (preferably linear) alkyl radical having 6 to 18 C atoms and Q₂ is a (preferably linear) alkyl radical having 4 to 16 C atoms.

Particularly preferred components of type (i) in the oil phase are as follows: isopropyl myristate, isopropyl palmitate, isopropyl stearate, isopropyl oleate, n-butyl stearate, n-hexyl laurate, n-decyl oleate, isooctyl stearate, isononyl stearate, isononyl isononanoate, 2-ethylhexyl palmitate, 2-ethylhexyl laurate, 2-hexyldecyl stearate, 2-octyldodecyl palmitate, oleyl oleate, oleyl erucate, erucyl oleate, erucyl erucate, 2-ethylhexyl isostearate, isotridecyl isononanoate, 2-ethylhexyl cocoate, caprylic/capric triglyceride, Alkylenediol dicaprylate caprates especially propylene-diol dicaprylate caprate; and also synthetic, semisynthetic and natural mixtures of such esters, e.g. jojoba oil.

Fatty acid triglycerides (oil components of type (i) in the oil phase) may also be in the form of, or in the form of a constituent of, synthetic, semisynthetic and/or natural oils, examples being olive oil, sunflower oil, soya oil, peanut oil, rapeseed oil, almond oil, palm oil, coconut oil, palm kernel oil and mixtures thereof.

Particularly preferred oil components of type (vii) in the oil phase are as follows: 2-butyl-1-octanol, 2-butyl-1-decanol, 2-octyl-1-dodecanol, 2-decyltetradecanol, 2-dodecyl-1-hexadecanol and 2-tetradecyl-1-octadecanol.

Particularly preferred oil components in the oil phase are mixtures comprising C₁₂-C₁₅-alkyl benzoate and 2-ethyl-

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hexyl isostearate, mixtures comprising C_{12} - C_{15} -alkyl benzoate and isotridecyl isononanoate, mixtures comprising C_{12} - C_{15} -alkyl benzoate, 2-ethylhexyl isostearate and isotridecyl isononanoate, mixtures comprising cyclomethicone and isotridecyl isononanoate, and mixtures comprising cyclomethicone and 2-ethylhexyl isostearate.

Preferred oil bodies, which form constituents of the O/W emulsions, are, for example, Guerbet alcohols based on fatty alcohols having 6 to 18, preferably 8 to 10, carbon atoms, esters of linear C_6 - C_{22} -fatty acids with linear or branched C_6 - C_{22} -fatty alcohols or esters of branched C_6 - C_{13} -carboxylic acids with linear or branched C_6 - C_{22} -fatty alcohols, such as, for example, myristyl myristate, myristyl palmitate, myristyl stearate, myristyl isostearate, myristyl oleate, myristyl behenate, myristyl erucate, cetyl myristate, cetyl palmitate, cetyl stearate, cetyl isostearate, cetyl oleate, cetyl behenate, cetyl erucate, stearyl myristate, stearyl palmitate, stearyl stearate, stearyl isostearate, stearyl oleate, stearyl behenate, stearyl erucate, isostearyl myristate, isostearyl palmitate, isostearyl stearate, isostearyl isostearate, isostearyl oleate, isostearyl behenate, isostearyl oleate, oleyl myristate, oleyl palmitate, oleyl stearate, oleyl isostearate, oleyl oleate, oleyl behenate, oleyl erucate, behenyl myristate, behenyl palmitate, behenyl stearate, behenyl isostearate, behenyl oleate, behenyl behenate, behenyl erucate, erucyl myristate, erucyl palmitate, erucyl stearate, erucyl isostearate, erucyl oleate, erucyl behenate and erucyl erucate. Also suitable are esters of linear C_6 - C_{22} -fatty acids with branched alcohols, in particular 2-ethylhexanol, esters of C_{18} - C_{38} -alkylhydroxy carboxylic acids with linear or branched C_6 - C_{22} -fatty alcohols, in particular Dioctyl Malate, esters of linear and/or branched fatty acids with polyhydric alcohols (such as, for example, propylene glycol, dimerdiol or trimertriol) and/or Guerbet alcohols, triglycerides based on C_6 - C_{10} -fatty acids, liquid mono-/di-/triglyceride mixtures based on C_6 - C_{13} -fatty acids, esters of C_6 - C_{22} -fatty alcohols and/or Guerbet alcohols with aromatic carboxylic acids, in particular benzoic acid, esters of C_2 - C_{12} -dicarboxylic acids with linear or branched alcohols having 1 to 22 carbon atoms or polyols having 2 to 10 carbon atoms and 2 to 6 hydroxyl groups, vegetable oils, branched primary alcohols, substituted cyclohexanes, linear and branched C_6 - C_{22} -fatty alcohol carbonates, such as, for example, Dicaprylyl Carbonate (Cetiol® CC), Guerbet carbonates, based on fatty alcohols having 6 to 18, preferably 8 to 10, carbon atoms, esters of benzoic acid with linear and/or branched C_6 - C_{22} -alcohols (e.g. Finsolv® TN), linear or branched, symmetrical or asymmetrical dialkyl ethers having 6 to 22 carbon atoms per alkyl group, such as, for example, dicaprylyl ether (Cetiol® OE), ring-opening products of epoxidized fatty acid esters with polyols, silicone oils (cyclomethicones, silicone methicone grades, etc.) and/or aliphatic or naphthenic hydrocarbons, such as, for example, squalane, squalene or dialkylcyclohexanes.

The most preferred oil components are triglycerides, particularly those of natural origin.

Actives

In a preferred embodiment the microcapsules of the present invention are loaded with one or more active, such as for example fragrances or perfume oils. For certain applications, also other additives can be used. With regard to fragrances and perfume oils said polyurea capsules allow loading high amounts up to 80 wt.-% calculated on the total capsule weight. The actives are preferably incorporated into the oil phase, however depending on their polarity they can also be incorporated into the first aqueous phase.

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Suitable fragrances and perfume oils are mixtures of natural and synthetic perfumes. Natural perfumes include the extracts of blossoms (lily, lavender, rose, jasmine, neroli, ylangylang), stems and leaves (geranium, patchouli, petit-grain), fruits (anise, coriander, caraway, juniper), fruit peel (bergamot, lemon, orange), roots (nutmeg, angelica, celery, cardamom, costus, iris, calmus), woods (pinewood, sandalwood, guaiac wood, cedarwood, rosewood), herbs and grasses (tarragon, lemon grass, sage, thyme), needles and branches (spruce, fir, pine, dwarf pine), resins and balsams (galbanum, elemi, benzoin, myrrh, olibanum, opoponax). Animal raw materials, for example civet and beaver, may also be used. Typical synthetic perfume compounds are products of the ester, ether, aldehyde, ketone, alcohol and hydrocarbon type. Examples of perfume compounds of the ester type are benzyl acetate, phenoxyethyl isobutyrate, p-tert.butyl cyclohexylacetate, linalyl acetate, dimethyl benzyl carbonyl acetate, phenyl ethyl acetate, linalyl benzoate, benzyl formate, ethylmethyl phenyl glycinate, allyl cyclohexyl propionate, styrallyl propionate and benzyl salicylate. Ethers include, for example, benzyl ethyl ether while aldehydes include, for example, the linear alkanals containing 8 to 18 carbon atoms, citral, citronellal, citronellyloxyacetaldehyde, cyclamen aldehyde, hydroxycitronellal, linal and bourgeonal. Examples of suitable ketones are the ionones, isomethylionone and methyl cedryl ketone. Suitable alcohols are anethol, citronellol, eugenol, isoeugenol, geraniol, linalool, phenylethyl alcohol and terpineol. The hydrocarbons mainly include the terpenes and balsams. However, it is preferred to use mixtures of different perfume compounds which, together, produce an agreeable perfume. Other suitable perfume oils are essential oils of relatively low volatility which are mostly used as aroma components. Examples are sage oil, camomile oil, clove oil, melissa oil, mint oil, cinnamon leaf oil, limeblossom oil, juniper berry oil, vetiver oil, olibanum oil, galbanum oil, ladanum oil and lavandin oil. The following are preferably used either individually or in the form of mixtures: bergamot oil, dihydro-myrcenol, linal, lylal, citronellol, phenylethyl alcohol, hexylcinnamaldehyde, geraniol, benzyl acetone, cyclamen aldehyde, linalool, Boisambrene Forte, Ambroxan, indole, hedione, sandelice, citrus oil, mandarin oil, orange oil, allylamyl glycolate, cyclovertal, lavandin oil, clary oil, damascone, geranium oil bourbon, cyclohexyl salicylate, Vertofix Coeur, Iso-E-Super, Fixolide NP, evermyl, iraldein gamma, phenylacetic acid, geranyl acetate, benzyl acetate, rose oxide, romillat, irotyl and floramat.

Colloidal Protection Agents

While the oil phase contains at least the isocyanate compound and optionally oil components and/or actives, the first aqueous phase contains the encapsulation aids, namely said colloidal protection agents which are preferably selected from the group consisting of chemically modified biopolymers, preferably chemically modified starches, modified gum Arabic, or modified cellulose. These are counted under the norm EI 450 and represent for examples starches (often derived from maize, quinoa, oats, waxy barley, or potato), gum Arabic, or cellulose which are chemically modified for example by octenyl succinic anhydride (OSA). Respective products are obtainable in the market for example CAPSUL™ STARCH or HI-CAP™ 100 (Ingredion Inc.). Typically, aqueous solutions comprising the aids in amounts of from 1 to 5 wt.-% are used.

Emulsifiers

In order to facilitate emulsification it can be useful adding non-interfering (non-agglomerating) emulsifiers selected from the group of non-ionic, anionic, amphoteric, and cat-

ionic surfactants and mixtures thereof to the first aqueous phase, which already contains the encapsulation aids.

Suitable non-ionic emulsifiers encompass for example:

products of the addition of 2 to 30 mol ethylene oxide and/or 0 to 5 mol propylene oxide onto linear C₈₋₂₂ fatty alcohols, onto C₁₂₋₂₂ fatty acids and onto alkyl phenols containing 8 to 15 carbon atoms in the alkyl group;

C_{12/18} fatty acid monoesters and diesters of addition products of 1 to 30 mol ethylene oxide onto glycerol; glycerol mono- and diesters and sorbitan mono- and diesters of saturated and unsaturated fatty acids containing 6 to 22 carbon atoms and ethylene oxide addition products thereof;

addition products of 15 to 60 mol ethylene oxide onto castor oil and/or hydrogenated castor oil;

polyol esters and, in particular, polyglycerol esters such as, for example, polyglycerol polyricinoleate, polyglycerol poly-12-hydroxystearate or polyglycerol dimerate isostearate. Mixtures of compounds from several of these classes are also suitable;

addition products of 2 to 15 mol ethylene oxide onto castor oil and/or hydrogenated castor oil;

partial esters based on linear, branched, unsaturated or saturated C₆₋₂₂ fatty acids, ricinoleic acid and 12-hydroxystearic acid and glycerol, polyglycerol, pentaerythritol, dipentaerythritol, sugar alcohols (for example sorbitol), alkyl glucosides (for example methyl glucoside, butyl glucoside, lauryl glucoside) and polyglucosides (for example cellulose);

mono-, di and trialkyl phosphates and mono-, di- and/or tri-PEG-alkyl phosphates and salts thereof;

wool wax alcohols;

polysiloxane/polyalkyl polyether copolymers and corresponding derivatives;

mixed esters of pentaerythritol, fatty acids, citric acid and fatty alcohol and/or mixed esters of C₆₋₂₂ fatty acids, methyl glucose and polyols, preferably glycerol or polyglycerol,

polyalkylene glycols and glycerol carbonate.

The addition products of ethylene oxide and/or propylene oxide onto fatty alcohols, fatty acids, alkylphenols, glycerol mono- and diesters and sorbitan mono- and diesters of fatty acids or onto castor oil are known commercially available products. They are homologue mixtures of which the average degree of alkoxylation corresponds to the ratio between the quantities of ethylene oxide and/or propylene oxide and substrate with which the addition reaction is carried out. C_{12/18} fatty acid monoesters and diesters of addition products of ethylene oxide onto glycerol are known as lipid layer enhancers for cosmetic formulations. The preferred emulsifiers are described in more detail as follows:

Partial glycerides. Typical examples of suitable partial glycerides are hydroxystearic acid monoglyceride, hydroxystearic acid diglyceride, isostearic acid monoglyceride, isostearic acid diglyceride, oleic acid monoglyceride, oleic acid diglyceride, ricinoleic acid monoglyceride, ricinoleic acid diglyceride, linoleic acid monoglyceride, linoleic acid diglyceride, linolenic acid monoglyceride, linolenic acid diglyceride, erucic acid monoglyceride, erucic acid diglyceride, tartaric acid monoglyceride, tartaric acid diglyceride, citric acid monoglyceride, citric acid diglyceride, malic acid monoglyceride, malic acid diglyceride and technical mixtures thereof which may still contain small quantities of triglyceride from the production process. Addition products

of 1 to 30 and preferably 5 to 10 mol ethylene oxide onto the partial glycerides mentioned are also suitable.

Sorbitan esters. Suitable sorbitan esters are sorbitan monoisostearate, sorbitan sesquiisostearate, sorbitan diisostearate, sorbitan triisostearate, sorbitan monooleate, sorbitan sesquioleate, sorbitan dioleate, sorbitan trioleate, sorbitan monoerucate, sorbitan sesquierucate, sorbitan dierucate, sorbitan trierucate, sorbitan monoricinoleate, sorbitan sesquiricinoleate, sorbitan diricinoleate, sorbitan triricinoleate, sorbitan monohydroxystearate, sorbitan sesquihydroxystearate, sorbitan dihydroxystearate, sorbitan trihydroxystearate, sorbitan monotartrate, sorbitan sesquitartrate, sorbitan ditartrate, sorbitan tritartrate, sorbitan monocitrate, sorbitan sesquicitrate, sorbitan dicitrate, sorbitan tricitrate, sorbitan monomaleate, sorbitan sesquimaleate, sorbitan dimaleate, sorbitan trimaleate and technical mixtures thereof. Addition products of 1 to 30 and preferably 5 to 10 mol ethylene oxide onto the sorbitan esters mentioned are also suitable.

Polyglycerol esters. Typical examples of suitable polyglycerol esters are Polyglyceryl-2 Dipolyhydroxystearate (Dehymuls® PGPH), Polyglycerin-3-Diisostearate (Lameform® TGI), Polyglyceryl-4 Isostearate (Isolan® GI 34), Polyglyceryl-3 Oleate, Diisostearoyl Polyglyceryl-3 Diisostearate (Isolan® PDI), Polyglyceryl-3 Methylglucose Distearate (Tego Care® 450), Polyglyceryl-3 Beeswax (Cera Bellina®), Polyglyceryl-4 Caprate (Polyglycerol Caprate T2010/90), Polyglyceryl-3 Cetyl Ether (Chime-xane® NL), Polyglyceryl-3 Distearate (Cremophor® GS 32) and Polyglyceryl Polyricinoleate (Admul® WOL 1403), Polyglyceryl Dimerate Isostearate and mixtures thereof. Examples of other suitable polyolesters are the mono-, di- and triesters of trimethylol propane or pentaerythritol with lauric acid, cocofatty acid, tallow fatty acid, palmitic acid, stearic acid, oleic acid, behenic acid and the like optionally reacted with 1 to 30 mol ethylene oxide.

Tetraalkyl ammonium salts. Cationically active surfactants comprise the hydrophobic high molecular group required for the surface activity in the cation by dissociation in aqueous solution. A group of important representatives of the cationic surfactants are the tetraalkyl ammonium salts of the general formula: (R¹R²R³R⁴N⁺)X⁻. Here R¹ stands for C₁-C₈ alk(en)yl, R², R³ and R⁴, independently of each other, for alk(en)yl radicals having 1 to 22 carbon atoms. X is a counter ion, preferably selected from the group of the halides, alkyl sulfates and alkyl carbonates. Cationic surfactants, in which the nitrogen group is substituted with two long acyl groups and two short alk(en)yl groups, are particularly preferred.

Esterquats. A further class of cationic surfactants particularly useful as co-surfactants for the present invention is represented by the so-called esterquats. Esterquats are generally understood to be quaternised fatty acid triethanolamine ester salts. These are known compounds which can be obtained by the relevant methods of preparative organic chemistry. Reference is made in this connection to International patent application WO 91/01295 A1, according to which triethanolamine is partly esterified with fatty acids in the presence of hypophosphorous acid, air is passed through the reaction mixture and the whole is then quaternised with dimethyl sulphate or ethylene oxide. In addition, German patent DE 4308794 C1 describes a process for the production of solid esterquats in which the quaternisation of triethanolamine esters is carried out in the presence of suitable dispersants, preferably fatty alcohols.

Typical examples of esterquats suitable for use in accordance with the invention are products of which the acyl

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component derives from monocarboxylic acids corresponding to formula RCOOH in which RCO is an acyl group containing 6 to 10 carbon atoms, and the amine component is triethanolamine (TEA). Examples of such monocarboxylic acids are caproic acid, caprylic acid, capric acid and technical mixtures thereof such as, for example, so-called head-fractionated fatty acid. Esterquats of which the acyl component derives from monocarboxylic acids containing 8 to 10 carbon atoms, are preferably used. Other esterquats are those of which the acyl component derives from dicarboxylic acids like malonic acid, succinic acid, maleic acid, fumaric acid, glutaric acid, sorbic acid, pimelic acid, azelaic acid, sebacic acid and/or dodecanedioic acid, but preferably adipic acid. Overall, esterquats of which the acyl component derives from mixtures of monocarboxylic acids containing 6 to 22 carbon atoms, and adipic acid are preferably used. The molar ratio of mono and dicarboxylic acids in the final esterquat may be in the range from 1:99 to 99:1 and is preferably in the range from 50:50 to 90:10 and more particularly in the range from 70:30 to 80:20. Besides the quaternised fatty acid triethanolamine ester salts, other suitable esterquats are quaternized ester salts of mono/dicarboxylic acid mixtures with diethanolalkylamines or 1,2-dihydroxypropyl dialkylamines. The esterquats may be obtained both from fatty acids and from the corresponding triglycerides in admixture with the corresponding dicarboxylic acids. One such process, which is intended to be representative of the relevant prior art, is proposed in European patent EP 0750606 B1. To produce the quaternised esters, the mixtures of mono- and dicarboxylic acids and the triethanolamine—based on the available carboxyl functions—may be used in a molar ratio of 1.1:1 to 3:1. With the performance properties of the esterquats in mind, a ratio of 1.2:1 to 2.2:1 and preferably 1.5:1 to 1.9:1 has proved to be particularly advantageous. The preferred esterquats are technical mixtures of mono-, di- and triesters with an average degree of esterification of 1.5 to 1.9.

The use of a combination of anionic and/or amphoteric surfactants with one or more nonionic surfactants is further advantageous. In a preferred embodiment according to the invention the composition further comprises emulsifiers that do not cause agglomeration in the required product formulations selected from the group consisting of:

- Alkyl phosphate derivatives
- Glyceryl oleate citrate derivatives
- Glyceryl stearate citrate derivatives
- Stearic acid esters
- Sorbitan esters
- Ethoxylated sorbitan esters
- Ethoxylated mono-, di- and tri glycerides
- Methyl glucose esters

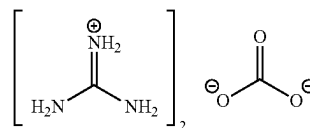
Emulsion and Dispersion

In a first step, an emulsion is prepared by blending the oil phase (comprising the isocyanate component and optionally oil components and/or actives) and the first aqueous phase (comprising the encapsulation aids and colloidal protecting agents). Preferably emulsification takes place by subjecting the mixture to high-speed shearing, for example using an Ultra-Turrax at 3,000 to 5,000 rpm over about 20 to about 120 seconds.

Once the emulsion is prepared it is blended with said second aqueous phase containing the guanidine carbonate that forms the polyurea by reaction of its amine groups with the isocyanate components. The mixture is also subjected to high-speed shearing under similar conditions as discussed above.

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Guanidine carbonate is a white, crystalline solid of the following formula



which is obtainable for example from Sigma-Aldrich. Said isocyanates and said guanidine carbonate are typically reacted in molar quantities of 1:1 to 5:1, preferably 1.2:1 to 3:1 calculated on the ratio between isocyanate and amine groups. In other words, a (small) excess of isocyanate is preferred.

The second aqueous solution may contain at least one surface active agent. Preferably said agent is a non-ionic and/or cationic polymer, as for example Suitable polymers to improve the spreadability of the composition upon the skin or hair, or improve the water and or sweat and or rub-off resistancy of the formula and to improve the protection factor of the composition. Examples of such polymers are: VP/Eicosene copolymers sold under the trade name of Antaron V-220 by International Speciality Products, VP/Hexadecene copolymer sold under the trade names Antaron V-216 and Antaron V-516 by International Speciality Products, Tricontanyl PVP sold under the trade name of Antaron WP-660 by International Speciality Products, Isohexadecane and Ethylene/Propylene/Styrene copolymer and Butylene/Styrene copolymer sold under the trade names of Versagel MC and MD by Penreco, Hydrogenated polyisobutene and Ethylene/Propylene/Styrene copolymer and Butylene/Styrene copolymer sold under the trade name of Versagel ME by Penreco, Acrylates/Octylacrylamide Copolymers sold under the trade name of Dermacryl 79, Dermacryl AQF and Dermacryl LT by AkzoNobel, Polyurethanes such as PPG-17/IPDI/DMPA copolymer sold under the trade name of Avalure UR 450 & 525 sold by Noveon, Polyurethanes-2 and -4 sold under the trade names Avalure UR-405, -410, 425, -430 and -445 525 sold by Noveon, Polyurethane 5 and Butyl Acetate and isopropyl alcohol sold under the trade name Avalure UR-510 and -525 sold by Noveon, Polyurethanes-1 and -6 sold under the trade name of Luviset PUR by BASF, Hydrogenated Dimer Dilinoleyl/Dimethylcarbonate Copolymer sold under the trade name of Cosmedia DC by Cognis.

Curing

Once the emulsion is blended with the second aqueous phase polyaddition takes place and the crude microcapsules incorporating the actives are formed. It is understood that the encapsulation aids are either incorporated into the capsule shell or bond to its surface. At that time the microcapsules show insufficient stability in the dispersion, therefore a final curing step is necessary. Typically curing takes place by treating the dispersion thus obtained at elevated temperatures of from about 50 to about 90° C. over a period of from about 1 to about 12 hours.

If desired the solvent can be removed to obtain the “pure” capsules, which typically show an average diameter of from about 5 to about 50 microns.

Liquid Surfactant Systems

Another object of the present invention refers to liquid surfactant systems, preferably detergent compositions com-

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prising either the microcapsules or the dispersion comprising the microcapsules as described above. Typically the amounts range from about 0.5 to about 5 wt.-percent and preferably from about 1 to about 2 wt.-percent—calculated as pure microcapsules on the total composition.

Suitable examples for liquid detergents encompass heavy duty liquid detergents, light duty liquid detergents, fabric softeners, manual dish wash agents, all-purpose cleaners and the like. Also included are anionic Ild type surfactant systems including body washes, most effective in systems with a viscosity range of 10-300 cpsec(est)

The detergent compositions according to the present invention may comprise any of the ingredients customarily found in such compositions, such as, for example, anionic, nonionic, cationic, amphoteric or zwitterionic (co-)surfactants, organic solvents, builders, enzymes and additional auxiliaries such as soil repellents, thickeners, colorants and fragrances or the like.

Anionic and Zwitterionic Co-Surfactants

Typical examples for anionic and zwitterionic surfactants encompass: Almondamidopropylamine Oxide, Almondamidopropyl Betaine, Aminopropyl Laurylglutamine, Ammonium C12-15 Alkyl Sulfate, Ammonium C12-16 Alkyl Sulfate, Ammonium Capryleth Sulfate, Ammonium Cocomonoglyceride Sulfate, Ammonium Coco-Sulfate, Ammonium Cocoyl Isethionate, Ammonium Cocoyl Sarcosinate, Ammonium C12-15 Pareth Sulfate, Ammonium C9-10 Perfluoroalkylsulfonate, Ammonium Dinonyl Sulfosuccinate, Ammonium Dodecylbenzenesulfonate, Ammonium Isostearate, Ammonium Laureth-6 Carboxylate, Ammonium Laureth-8 Carboxylate, Ammonium Laureth Sulfate, Ammonium Laureth-5 Sulfate, Ammonium Laureth-7 Sulfate, Ammonium Laureth-9 Sulfate, Ammonium Laureth-12 Sulfate, Ammonium Lauroyl Sarcosinate, Ammonium Lauryl Sulfate, Ammonium Lauryl Sulfosuccinate, Ammonium Myreth Sulfate, Ammonium Myristyl Sulfate, Ammonium Nonoxynol-4 Sulfate, Ammonium Nonoxynol-30 Sulfate, Ammonium Oleate, Ammonium Palm Kernel Sulfate, Ammonium Stearate, Ammonium Tallate, AMPD-Isostearoyl Hydrolyzed Collagen, AMPD-Rosin Hydrolyzed Collagen, AMP-Isostearoyl Hydrolyzed Collagen, AMP-Isostearoyl Hydrolyzed Keratin, AMP-Isostearoyl Hydrolyzed Soy Protein, AMP-Isostearoyl Hydrolyzed Wheat Protein, Apricotamidopropyl Betaine, Arachidic Acid, Arginine Hexyldecyl Phosphate, Avocamidopropyl Betaine, Avocado Oil Glycereth-8 Esters, Babassu Acid, Babassuamidopropylamine Oxide, Babassuamidopropyl Betaine, Beeswax Acid, Behenamidopropyl Betaine, Behenamine Oxide, Beheneth-25, Beheneth-30, Behenic Acid, Behenyl Betaine, Bis-Butyldimethicone Polyglyceryl-3, Butoxynol-5 Carboxylic Acid, Butoxynol-19 Carboxylic Acid, Butyldimoniumhydroxypropyl Butylglucosides Chloride, Butyldimoniumhydroxypropyl Laurylglucosides Chloride, Butyl Glucoside, Butylglucoside Caprate, Butylglucosides Hydroxypropyltrimonium Chloride, Butyloctanoic Acid, C18-36 Acid, C20-40 Acid, C30-50 Acid, C16-22 Acid Amide MEA, Calcium Dodecylbenzenesulfonate, Calcium Lauroyl Taurate, C9-16 Alkane/Cycloalkane, C10-14 Alkyl Benzenesulfonic Acid, C12-14 Alkyl Diaminoethylglycine HCL, C9-15 Alkyl Phosphate, *Candida Bombicola*/Glucose/Methyl Rapeseedate Ferment, Canolamidopropyl Betaine, Capric Acid, Caproic Acid, Caproyl Ethyl Glucoside, Capryl/Capramidopropyl Betaine, Capryleth-4 Carboxylic Acid, Capryleth-6 Carboxylic Acid, Capryleth-9 Carboxylic Acid, Caprylic Acid, Capryloyl Collagen Amino Acids, Capryloyl Glycine, Capryloyl Hydrolyzed Collagen, Capryloyl Hydrolyzed Keratin, Capryloyl

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Keratin Amino Acids, Capryloyl Silk Amino Acids, Capryloyl/Capryl Glucoside, Capryloyl/Capryl Wheat Bran/Straw Glycosides, Capryloyl Glucoside, Capryloyl Glyceryl Ether, Capryloyl Pyrrolidone, Carnitine, Cetareth-20, Cetareth-23, Cetareth-24, Cetareth-25, Cetareth-27, Cetareth-28, Cetareth-29, Cetareth-30, Cetareth-33, Cetareth-34, Cetareth-40, Cetareth-50, Cetareth-55, Cetareth-60, Cetareth-80, Cetareth-100, Cetareth-25 Carboxylic Acid, Cetareth-2 Phosphate, Cetareth-4 Phosphate, Cetareth-5 Phosphate, Cetareth-10 Phosphate, Ceteth-20, Ceteth-23, Ceteth-24, Ceteth-25, Ceteth-30, Ceteth-40, Ceteth-45, Ceteth-150, Ceteth-8 Phosphate, Ceteth-10 Phosphate, Ceteth-20 Phosphate, Cetoeth-22, Cetoeth-24, Cetoeth-25, Cetoeth-30, Cetyl Betaine, *Chrysanthemum Sinense* Flower Extract, C12-14 Hydroxyalkyl Hydroxyethyl Beta-Alanine, C12-14 Hydroxyalkyl Hydroxyethyl Sarcosine, Cocamidethyl Betaine, Cocamidopropylamine Oxide, Cocamidopropyl Betainamide MEA Chloride, Cocamidopropyl Betaine, Cocamidopropyl Hydroxysultaine, Cocamine Oxide, Cocaminobutyric Acid, Cocaminopropionic Acid, Coceth-7 Carboxylic Acid, Coceth-4 Glucoside, Cocoamphodipropionic Acid, Cocobetainamido Amphopropionate, Cocobetaine, Cocodimonium Hydroxypropyl Hydrolyzed Rice Protein, Cocodimonium Hydroxypropyl Hydrolyzed Soy Protein, Cocodimonium Hydroxypropyl Hydrolyzed Wheat Protein, CocoGlucoside, Cocoglucosides Hydroxypropyltrimonium Chloride, Coco-Hydroxysultaine, CocoMorpholine Oxide, Coconut Acid, Coconut Oil Glycereth-8 Esters, Coco/Oleamidopropyl Betaine, Coco-Sultaine, Coco/Sunfloweramidopropyl Betaine, Cocoylcholine Methosulfate, Cocoyl Glutamic Acid, Cocoyl Hydrolyzed Collagen, Cocoyl Hydrolyzed Keratin, Cocoyl Hydrolyzed Oat Protein, Cocoyl Hydrolyzed Rice Protein, Cocoyl Hydrolyzed Silk, Cocoyl Hydrolyzed Soy Protein, Cocoyl Hydrolyzed Wheat Protein, Cocoyl Sarcosine, Corn Acid, Cottonseed Acid, Cottonseed Oil Glycereth-8 Esters, C10-16 Pareth-1, C10-16 Pareth-2, C11-13 Pareth-6, C11-13 Pareth-9, C11-13 Pareth-10, C11-15 Pareth-30, C11-15 Pareth-40, C12-13 Pareth-1, C12-13 Pareth-23, C12-14 Pareth-5, C12-14 Pareth-9, C13-15 Pareth-21, C14-15 Pareth-8, C20-22 Pareth-30, C20-40 Pareth-40, C20-40 Pareth-95, C22-24 Pareth-33, C30-50 Pareth-40, C9-11 Pareth-6 Carboxylic Acid, C9-11 Pareth-8 Carboxylic Acid, C11-15 Pareth-7 Carboxylic Acid, C12-13 Pareth-5 Carboxylic Acid, C12-13 Pareth-7 Carboxylic Acid, C12-13 Pareth-8 Carboxylic Acid, C12-13 Pareth-12 Carboxylic Acid, C12-15 Pareth-7 Carboxylic Acid, C12-15 Pareth-8 Carboxylic Acid, C12-15 Pareth-12 Carboxylic Acid, C14-15 Pareth-8 Carboxylic Acid, C6-10 Pareth-4 Phosphate, C12-13 Pareth-2 Phosphate, C12-13 Pareth-10 Phosphate, C12-15 Pareth-6 Phosphate, C12-15 Pareth-8 Phosphate, C12-15 Pareth-10 Phosphate, C12-16 Pareth-6 Phosphate, C4-18 Perfluoroalkylethyl Thiohydroxypropyltrimonium Chloride, Cupuassuamidopropyl Betaine, DEA-C12-13 Alkyl Sulfate, DEA-C12-15 Alkyl Sulfate, DEA-Cetareth-2 Phosphate, DEA-Cetyl Sulfate, DEA-Cocoamphodipropionate, DEA-C12-13 Pareth-3 Sulfate, DEA-Cyclocarboxypropyloleate, DEA-Dodecylbenzenesulfonate, DEA-Isostearate, DEA-Laureth Sulfate, DEALauryl Sulfate, DEA-Linoleate, DEA-Methyl Myristate Sulfonate, DEA-Myreth Sulfate, DEAMyristate, DEA-Myristyl Sulfate, DEA-Oleth-5 Phosphate, DEA-Oleth-20 Phosphate, DEA PGOleate, Deceth-7 Carboxylic Acid, Deceth-7 Glucoside, Deceth-9 Phosphate, Decylamine Oxide, Decyl Betaine, Decyl Glucoside, Decyltetradeceth-30, Decyltetradecylamine Oxide, Diammonium Lauramido-MEA Sulfosuccinate, Diammonium Lauryl Sulfosuccinate, Diammonium Oleamido PEG-2

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Sulfosuccinate, Dibutoxymethane, Di-CI 2-15 Pareth-2 Phosphate, Di-CI 2-15 Pareth-4 Phosphate, Di-CI 2-15 Pareth-6 Phosphate, Di-CI 2-15 Pareth-8 Phosphate, Di-CI 2-15 Pareth-10 Phosphate, Didodecyl Butanetetracarboxylate, Diethylamine Laureth Sulfate, Diethylhexyl Sodium Sulfosuccinate, Dihydroxyethyl C8-10 Alkoxypropylamine Oxide, Dihydroxyethyl C9-11 Alkoxypropylamine Oxide, Dihydroxyethyl C12-15 Alkoxypropylamine Oxide, Dihydroxyethyl Cocamine Oxide, Dihydroxyethyl Lauramine Oxide, Dihydroxyethyl Stearamine Oxide, Dihydroxyethyl Tallowamine Oxide, Dimethicone PEG-7 Phosphate, Dimethicone PEG-10 Phosphate, Dimethicone PEG/PPG-7/4 Phosphate, Dimethicone PEG/PPG-12/4 Phosphate, Dimethicone/Polyglycerin-3 Crosspolymer, Dimethicone Propyl PG-Betaine, Dimyristyl Phosphate, Dioleoylamidoethyl Hydroxyethylmonium Methosulfate, DIPA-Hydrogenated Cocoate, DIPA-Lanolate, DIPA-Myristate, Dipotassium Capryloyl Glutamate, Dipotassium Lauryl Sulfosuccinate, Dipotassium Undecylenoyl Glutamate, Disodium Babassuamido MEA-Sulfosuccinate, Disodium Caproamphodiacetate, Disodium Caproamphodipropionate, Disodium Capryloamphodiacetate, Disodium Capryloamphodipropionate, Disodium Capryloyl Glutamate, Disodium Cetearyl Sulfosuccinate, Disodium Cetyl Phenyl Ether Disulfonate, Disodium Cetyl Sulfosuccinate, Disodium Cocamido MEA-Sulfosuccinate, Disodium Cocamido MIPA PEG-4 Sulfosuccinate, Disodium Cocamido MIPA-Sulfosuccinate, Disodium Cocamido PEG-3 Sulfosuccinate, Disodium Coceth-3 Sulfosuccinate, Disodium Cocoamphocarboxyethylhydroxypropylsulfonate, Disodium Cocoamphodiacetate, Disodium Cocoamphodipropionate, Disodium Coco-Glucoside Sulfosuccinate, Disodium Coco-Sulfosuccinate, Disodium Cocoyl Butyl Gluceth-10 Sulfosuccinate, Disodium Cocoyl Glutamate, Disodium C12-14 Pareth-1 Sulfosuccinate, Disodium C12-14 Pareth-2 Sulfosuccinate, Disodium C12-15 Pareth Sulfosuccinate, Disodium C12-14 Sec-Pareth-3 Sulfosuccinate, Disodium C12-14 Sec-Pareth-5 Sulfosuccinate, Disodium C12-14 Sec-Pareth-7 Sulfosuccinate, Disodium C12-14 Sec-Pareth-9 Sulfosuccinate, Disodium C12-14 Sec-Pareth-12 Sulfosuccinate, Disodium Deceth-5 Sulfosuccinate, Disodium Deceth-6 Sulfosuccinate, Disodium Decyl Phenyl Ether Disulfonate, Disodium Dihydroxyethyl Sulfosuccinylundecylenate, Disodium Ethylene Dicocamide PEG-15 Disulfate, Disodium Hydrogenated Cottonseed Glyceride Sulfosuccinate, Disodium Hydrogenated Tallow Glutamate, Disodium Hydroxydecyl Sorbitol Citrate, Disodium Isodecyl Sulfosuccinate, Disodium Isostearamido MEA-Sulfosuccinate, Disodium Isostearamido MIPA-Sulfosuccinate, Disodium Isostearoamphodiacetate, Disodium Isostearoamphodipropionate, Disodium Isostearyl Sulfosuccinate, Disodium Laneth-5 Sulfosuccinate, Disodium Lauramido MEA-Sulfosuccinate, Disodium Lauramido MIPA Glycol Sulfosuccinate, Disodium Lauramido PEG-2 Sulfosuccinate, Disodium Lauramido PEG-5 Sulfosuccinate, Disodium Laureth-5 Carboxyamphodiacetate, Disodium Laureth-7 Citrate, Disodium Laureth Sulfosuccinate, Disodium Laureth-6 Sulfosuccinate, Disodium Laureth-9 Sulfosuccinate, Disodium Laureth-12 Sulfosuccinate, Disodium Lauriminobishydroxypropylsulfonate, Disodium Lauriminodiacetate, Disodium Lauriminodipropionate, Disodium Lauriminodipropionate Tocopheryl Phosphates, Disodium Lauroamphodiacetate, Disodium Lauroamphodipropionate, Disodium N-Lauroyl Aspartate, Disodium Lauroyl Glutamate, Disodium Lauryl Phenyl Ether Disulfonate, Disodium Lauryl Sulfosuccinate, Disodium Myristamido MEA-Sulfosuccinate, Disodium Nonoxynol-10 Sulfosuccinate, Disodium Oleamido MEA-Sulfo-

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succinate, Disodium Oleamido MIPA-Sulfosuccinate, Disodium Oleamido PEG-2 Sulfosuccinate, Disodium Oleoamphodipropionate, Disodium Oleth-3 Sulfosuccinate, Disodium Oleyl Phosphate, Disodium Oleyl Sulfosuccinate, Disodium Palmitamido PEG-2 Sulfosuccinate, Disodium Palmitoleamido PEG-2 Sulfosuccinate, Disodium PEG-4 Cocamido MIPA-Sulfosuccinate, Disodium PEG-12 Dimethicone Sulfosuccinate, Disodium PEG-8 Palm Glycerides Sulfosuccinate, Disodium PPG-2-Isodeceth-7 Carboxyamphodiacetate, Disodium Ricinoleamido MEA-Sulfosuccinate, Disodium Sitostereth-14 Sulfosuccinate, Disodium Soyamphodiacetate, Disodium Stearamido MEA-Sulfosuccinate, Disodium Steariminodipropionate, Disodium Stearoamphodiacetate, Disodium Stearoyl Glutamate, Disodium Stearyl Sulfosuccinamate, Disodium Stearyl Sulfosuccinate, Disodium 2-Sulfolaurate, Disodium 2-Sulfopalmitate, Disodium Tallamido MEA-Sulfosuccinate, Disodium Tallowamido MEA-Sulfosuccinate, Disodium Tallowamphodiacetate, Disodium Tallowiminodipropionate, Disodium Tallow Sulfosuccinamate, Disodium Tridecylsulfosuccinate, Disodium Undecylenamido MEA-Sulfosuccinate, Disodium Undecylenamido PEG-2 Sulfosuccinate, Disodium Undecylenoyl Glutamate, Disodium Wheat Germamido MEA-Sulfosuccinate, Disodium Wheat Germamido PEG-2 Sulfosuccinate, Disodium Wheatgermamphodiacetate, Di-TEA-Cocamide Diacetate, Di-TEA-Oleamido PEG-2 Sulfosuccinate, Di-TEA-Palmitoyl Aspartate, Ditridecyl Sodium Sulfosuccinate, Dodecylbenzene Sulfonic Acid, Erucamidopropyl Hydroxysultaine, Ethylhexeth-3 Carboxylic Acid, Ethyl PEG-15 Cocamine Sulfate, Glyceryl Capryl Ether, Hexyldecanoic Acid, Hydrogenated Coconut Acid, Hydrogenated Laneth-25, Hydrogenated Menhaden Acid, Hydrogenated Palm Acid, Hydrogenated Palm Kernel Amine Oxide, Hydrogenated Tallow Acid, Hydrogenated Tallowamine Oxide, Hydrogenated Tallow Betaine, Hydrogenated Talloweth-25, Hydrogenated Tallowoyl Glutamic Acid, Hydrolyzed *Candida Bombicola* Extract, Hydroxyceteth-60, Hydroxyethyl Acetomonium PG-Dimethicone, Hydroxyethylbutylamine Laureth Sulfate, Hydroxyethyl Carboxymethyl Cocamidopropylamine, Hydroxyethyl Hydroxypropyl C12-15 Alkoxypropylamine Oxide, Hydroxylauryl/Hydroxymyristyl Betaine, Hydroxystearic Acid, Hydroxysuccinimidyl C10-40 Isoalkyl Acidate, Hydroxysuccinimidyl C21-22 Isoalkyl Acidate, Hydroxysultaines, IPDI/PEG-15 Soyamine Oxide Copolymer, IPDI/PEG-15 Soyethonium Ethosulfate Copolymer, IPDI/PEG-15 Soy Glycinate Copolymer, Isoceteth-30, Isolaurate-4 Phosphate, Isopolyglyceryl-3 Dimethicone, Isopolyglyceryl-3 Dimethiconol, Isopropanolamine Lanolate, Isopropylamine Dodecylbenzenesulfonate, Isostearamidopropylamine Oxide, Isostearamidopropyl Betaine, Isostearamidopropyl Morpholine Oxide, Isosteareth-8, Isosteareth-16, Isosteareth-22, Isosteareth-25, Isosteareth-50, Isostearic Acid, Isostearoyl Hydrolyzed Collagen, Jojoba Oil PEG-150 Esters, Jojoba Wax PEG-80 Esters, Jojoba Wax PEG-120 Esters, Laneth-20, Laneth-25, Laneth-40, Laneth-50, Laneth-60, Laneth-75, Lanolin Acid, Lauramidopropylamine Oxide, Lauramidopropyl Betaine, Lauramidopropyl Hydroxysultaine, Lauramine Oxide, Lauraminopropionic Acid, Laurdimoniumhydroxypropyl Decylglucosides Chloride, Laurdimoniumhydroxypropyl Laurylglucosides Chloride, Laureth-16, Laureth-20, Laureth-21, Laureth-23, Laureth-25, Laureth-30, Laureth-38, Laureth-40, Laureth-3 Carboxylic Acid, Laureth-4 Carboxylic Acid, Laureth-5 Carboxylic Acid, Laureth-6 Carboxylic Acid, Laureth-8 Carboxylic Acid, Laureth-10 Carboxylic Acid, Laureth-11 Carboxylic Acid, Laureth-12 Carboxylic

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Acid, Laureth-13 Carboxylic Acid, Laureth-14 Carboxylic Acid, Laureth-17 Carboxylic Acid, Laureth-6 Citrate, Laureth-7 Citrate, Laureth-1 Phosphate, Laureth-2 Phosphate, Laureth-3 Phosphate, Laureth-4 Phosphate, Laureth-7 Phosphate, Laureth-8 Phosphate, Laureth-7 Tartrate, Lauric Acid, Laurimino Bispropanediol, Lauriminodipropionic Acid, Lauroamphodipropionic Acid, Lauroyl Beta-Alanine, Lauroyl Collagen Amino Acids, Lauroyl Ethyltrimonium Methosulfate, Lauroyl Hydrolyzed Collagen, Lauroyl Hydrolyzed Elastin, Lauroyl Methyl Glucamide, Lauroyl Sarcosine, Lauroyl Silk Amino Acids, Lauryl Betaine, Lauryl Dimethicone/Polyglycerin-3 Crosspolymer, Lauryldimoniumhydroxypropyl Cocoglucosides Chloride, Lauryl Glucoside, Laurylglucosides Hydroxypropyltrimonium Chloride, Lauryl Glycol Hydroxypropyl Ether, Lauryl Hydroxysultaine, Lauryl Malamide, Lauryl Methylglucamide, Lauryl/Myristyl Glycol Hydroxypropyl Ether, Lauryl/Myristyl Wheat Bran/Straw Glycosides, Lauryl Polyglyceryl-3 Polydimethylsiloxylethyl Dimethicone, Lauryl Pyrrolidone, Lauryl Sultaine, Linoleic Acid, Linolenic Acid, Linseed Acid, Lysine Cocoate, Macadamia Seed Oil Glycereth-8 Esters, Magnesium Coceth Sulfate, Magnesium Coco-Sulfate, Magnesium Isododecylbenzenesulfonate, Magnesium Laureth-11 Carboxylate, Magnesium Laureth Sulfate, Magnesium Laureth-5 Sulfate, Magnesium Laureth-8 Sulfate, Magnesium Laureth-16 Sulfate, Magnesium Laureth-3 Sulfosuccinate, Magnesium Lauryl Hydroxypropyl Sulfonate, Magnesium Lauryl Sulfate, Magnesium Methyl Cocoyl Taurate, Magnesium Myreth Sulfate, Magnesium Oleth Sulfate, Magnesium/TEA-Coco-Sulfate, Manicouagan Clay, MEACocoate, MEA-Laureth-6 Carboxylate, MEA-Laureth Sulfate, MEA-Lauryl Sulfate, MEA PPG-6 Laureth-7 Carboxylate, MEA-PPG-8-Steareth-7 Carboxylate, MEA-Undecylenate, Meroxapol 108, Meroxapol 174, Meroxapol 178, Meroxapol 254, Meroxapol 255, Meroxapol 258, Meroxapol 314, Methoxy PEG-450 Amidoglutaryl Succinimide, Methoxy PEG-450 Amido Hydroxysuccinimidyl Succinamate, Methoxy PEG-450 Maleimide, Methyl Morpholine Oxide, Milkamidopropyl Amine Oxide, Milkamidopropyl Betaine, Minkamidopropylamine Oxide, Minkamidopropyl Betaine, MIPA C12-15 Pareth Sulfate, MIPA-Dodecylbenzenesulfonate, MIPA-Laureth Sulfate, MIPA-Lauryl Sulfate, Mixed Isopropanolamines Lanolate, Mixed Isopropanolamines Lauryl Sulfate, Mixed Isopropanolamines Myristate, Morpholine Oleate, Morpholine Stearate, Myreth-3 Carboxylic Acid, Myreth-5 Carboxylic Acid, Myristalkonium Chloride, Myristamidopropylamine Oxide, Myristamidopropyl Betaine, Myristamidopropyl Dimethylamine Phosphate, Myristamidopropyl Hydroxysultaine, Myristamidopropyl PG-Dimonium Chloride Phosphate, Myristamine Oxide, Myristaminopropionic Acid, Myristic Acid, Myristoyl Ethyltrimonium Methosulfate, Myristoyl Glutamic Acid, Myristoyl Hydrolyzed Collagen, Myristoyl Sarcosine, Myristyl Betaine, Myristyl/Cetyl Amine Oxide, Myristyldimoniumhydroxypropyl Cocoglucosides Chloride, Myristyl Glucoside, Myristyl Phosphate, Nonoxynol-20, Nonoxynol-23, Nonoxynol-25, Nonoxynol-30, Nonoxynol-35, Nonoxynol-40, Nonoxynol-44, Nonoxynol-50, Nonoxynol-100, Nonoxynol-120, Nonoxynol-5 Carboxylic Acid, Nonoxynol-8 Carboxylic Acid, Nonoxynol-10 Carboxylic Acid, Nonoxynol-3 Phosphate, Nonoxynol-4 Phosphate, Nonoxynol-6 Phosphate, Nonoxynol-9 Phosphate, Nonoxynol-10 Phosphate, Nonyl Nonoxynol-30, Nonyl Nonoxynol-49, Nonyl Nonoxynol-100, Nonyl Nonoxynol-150, Nonyl Nonoxynol-7 Phosphate, Nonyl Nonoxynol-8 Phosphate, Nonyl Nonoxynol-9 Phosphate, Nonyl Nonoxynol-10 Phosphate, Nonyl

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Nonoxynol-11 Phosphate, Nonyl Nonoxynol-15 Phosphate, Nonyl Nonoxynol-24 Phosphate, Oatamidopropyl Betaine, Octoxynol-16, Octoxynol-25, Octoxynol-30, Octoxynol-33, Octoxynol-40, Octoxynol-70, Octoxynol-20 Carboxylic Acid, Octyldodeceth-20, Octyldodeceth-25, Octyldodeceth-30, Oleamidopropylamine Oxide, Oleamidopropyl Betaine, Oleamidopropyl Hydroxysultaine, Oleamine Oxide, Oleic Acid, Oleoyl Hydrolyzed Collagen, Oleoyl Sarcosine, Oleth-20, Oleth-23, Oleth-24, Oleth-25, Oleth-30, Oleth-35, Oleth-40, Oleth-44, Oleth-50, Oleth-3 Carboxylic Acid, Oleth-6 Carboxylic Acid, Oleth-10 Carboxylic Acid, Oleyl Betaine, Olivamidopropylamine Oxide, Olivamidopropyl Betaine, Olive Acid, Olivoyl Hydrolyzed Wheat Protein, Ophiopogon Extract Stearate, Ozonized Oleth-10, Ozonized PEG-10 Oleate, Ozonized PEG-14 Oleate, Ozonized Polysorbate 80, Palm Acid, Palmamidopropyl Betaine, Palmeth-2 Phosphate, Palmitamidopropylamine Oxide, Palmitamidopropyl Betaine, Palmitamine Oxide, Palmitic Acid, Palmitoyl Collagen Amino Acids, Palmitoyl Glycine, Palmitoyl Hydrolyzed Collagen, Palmitoyl Hydrolyzed Milk Protein, Palmitoyl Hydrolyzed Wheat Protein, Palmitoyl Keratin Amino Acids, Palmitoyl Oligopeptide, Palmitoyl Silk Amino Acids, Palm Kernel Acid, Palm Kernelamidopropyl Betaine, Peach Kernel Oil Glycereth-8 Esters, Peanut Acid, PEG-10 Castor Oil, PEG-40 Castor Oil, PEG-44 Castor Oil, PEG-50 Castor Oil, PEG-54 Castor Oil, PEG-55 Castor Oil, PEG-60 Castor Oil, PEG-80 Castor Oil, PEG-100 Castor Oil, PEG-200 Castor Oil, PEG-11 Cocamide, PEG-6 Cocamide Phosphate, PEG-4 Cocamine, PEG-8 Cocamine, PEG-12 Cocamine, PEG-150 Dibehenate, PEG-90 Diisostearate, PEG-75 Dilaurate, PEG-150 Dilaurate, PEG-75 Dioleate, PEG-150 Dioleate, PEG-75 Distearate, PEG-120 Distearate, PEG-150 Distearate, PEG-175 Distearate, PEG-190 Distearate, PEG-250 Distearate, PEG-30 Glyceryl Cocoate, PEG-40 Glyceryl Cocoate, PEG-78 Glyceryl Cocoate, PEG-80 Glyceryl Cocoate, PEG-30 Glyceryl Isostearate, PEG-40 Glyceryl Isostearate, PEG-50 Glyceryl Isostearate, PEG-60 Glyceryl Isostearate, PEG-90 Glyceryl Isostearate, PEG-23 Glyceryl Laurate, PEG-30 Glyceryl Laurate, PEG-25 Glyceryl Oleate, PEG-30 Glyceryl Oleate, PEG-30 Glyceryl Soyate, PEG-25 Glyceryl Stearate, PEG-30 Glyceryl Stearate, PEG-40 Glyceryl Stearate, PEG-120 Glyceryl Stearate, PEG-200 Glyceryl Stearate, PEG-28 Glyceryl Tallowate, PEG-80 Glyceryl Tallowate, PEG-82 Glyceryl Tallowate, PEG-130 Glyceryl Tallowate, PEG-200 Glyceryl Tallowate, PEG-45 Hydrogenated Castor Oil, PEG-50 Hydrogenated Castor Oil, PEG-54 Hydrogenated Castor Oil, PEG-55 Hydrogenated Castor Oil, PEG-60 Hydrogenated Castor Oil, PEG-80 Hydrogenated Castor Oil, PEG-100 Hydrogenated Castor Oil, PEG-200 Hydrogenated Castor Oil, PEG-30 Hydrogenated Lanolin, PEG-70 Hydrogenated Lanolin, PEG-50 Hydrogenated Palmamide, PEG-2 Isostearate, PEG-3 Isostearate, PEG-4 Isostearate, PEG-6 Isostearate, PEG-8 Isostearate, PEG-10 Isostearate, PEG-12 Isostearate, PEG-20 Isostearate, PEG-30 Isostearate, PEG-40 Isostearate, PEG-26 Jojoba Acid, PEG-40 Jojoba Acid, PEG-15 Jojoba Alcohol, PEG-26 Jojoba Alcohol, PEG-40 Jojoba Alcohol, PEG-35 Lanolin, PEG-40 Lanolin, PEG-50 Lanolin, PEG-55 Lanolin, PEG-60 Lanolin, PEG-70 Lanolin, PEG-75 Lanolin, PEG-85 Lanolin, PEG-100 Lanolin, PEG-150 Lanolin, PEG-75 Lanolin Oil, PEG-2 Lauramide, PEG-3 Lauramine Oxide, PEG-20 Laurate, PEG-32 Laurate, PEG-75 Laurate, PEG-150 Laurate, PEG-70 Mango Glycerides, PEG-20 Mannitan Laurate, PEG-8 Methyl Ether Dimethicone, PEG-120 Methyl Glucose Dioleate, PEG-80 Methyl Glucose Laurate, PEG-120 Methyl Glucose Trioleate, PEG-4 Montanate, PEG-30

Oleamine, PEG-20 Oleate, PEG-23 Oleate, PEG-32 Oleate, PEG-36 Oleate, PEG-75 Oleate, PEG-150 Oleate, PEG-20 Palmitate, PEG-150 Polyglyceryl-2 Tristearate, PEG/PPG-28/21 Acetate Dimethicone, PEG/PPG-24/18 Butyl Ether Dimethicone, PEG/PPG-3/17 Copolymer, PEG/PPG-5/35 Copolymer, PEG/PPG-8/55 Copolymer, PEG/PPG-10/30 Copolymer, PEG/PPG-10/65 Copolymer, PEG/PPG-12/35 Copolymer, PEG/PPG-16/17 Copolymer, PEG/PPG-20/9 Copolymer, PEG/PPG-20/20 Copolymer, PEG/PPG-20/60 Copolymer, PEG/PPG-20/65 Copolymer, PEG/PPG-22/25 Copolymer, PEG/PPG-28/30 Copolymer, PEG/PPG-30-35 Copolymer, PEG/PPG-30/55 Copolymer, PEG/PPG-35/40 Copolymer, PEG/PPG-50/40 Copolymer, PEG/PPG-150/35 Copolymer, PEG/PPG-160/30 Copolymer, PEG/PPG-190/60 Copolymer, PEG/PPG-200/40 Copolymer, PEG/PPG-300/55 Copolymer, PEG-20 Sorbitan Isostearate, PEG-40 Sorbitan Lanolate, PEG-75 Sorbitan Lanolate, PEG-10 Sorbitan Laurate, PEG-40 Sorbitan Laurate, PEG-44 Sorbitan Laurate, PEG-75 Sorbitan Laurate, PEG-80 Sorbitan Laurate, PEG-20 Sorbitan Oleate, PEG-80 Sorbitan Palmitate, PEG-40 Sorbitan Stearate, PEG-60 Sorbitan Stearate, PEG-160 Sorbitan Triisostearate, PEG-40 Soy Sterol, PEG-2 Stearamide Carboxylic Acid, PEG-9 Stearamide Carboxylic Acid, PEG-20 Stearate, PEG-23 Stearate, PEG-25 Stearate, PEG-30 Stearate, PEG-32 Stearate, PEG-35 Stearate, PEG-36 Stearate, PEG-40 Stearate, PEG-45 Stearate, PEG-50 Stearate, PEG-55 Stearate, PEG-75 Stearate, PEG-90 Stearate, PEG-100 Stearate, PEG-120 Stearate, PEG-150 Stearate, PEG-45 Stearate Phosphate, PEG-20 Tallate, PEG-50 Tallow Amide, PEG-2 Tallowamide DEA, PEG-20 Tallowate, PEG-66 Trihydroxystearin, PEG-200 Trihydroxystearin, PEG-60 Tsubakiate Glycerides, Pelargonic Acid, Pentadecynol-200, Pheneth-6 Phosphate, Poloxamer 105, Poloxamer 108, Poloxamer 182, Poloxamer 183, Poloxamer 184, Poloxamer 188, Poloxamer 217, Poloxamer 234, Poloxamer 235, Poloxamer 237, Poloxamer 238, Poloxamer 288, Poloxamer 334, Poloxamer 335, Poloxamer 338, Poloxamine 908, Poloxamine 1508, Polydimethylsiloxo PEG/PPG-24/19 Butyl Ether Silsesquioxane, Polydimethylsiloxo PPG-13 Butyl Ether Silsesquioxane, Polyglyceryl-6 Caprate, Polyglyceryl-10 Dilaurate, Polyglyceryl-20 Heptacaprilate, Polyglyceryl-20 Hexacaprilate, Polyglyceryl-2 Lauryl Ether, Polyglyceryl-10 Lauryl Ether, Polyglyceryl-20 Octaizononanoate, Polyglyceryl-6 Pentacaprilate, Polyglyceryl-10 Pentacaprilate, Polyglyceryl-3 Polydimethylsiloxoethyl Dimethicone, Polyglyceryl-6 Tetracaprilate, Polyglyceryl-10 Tetralaurate, Polyglyceryl-6 Tricaprilate, Polyglyceryl-10 Trilaurate, Polyquaternium-77, Polyquaternium-78, Polyquaternium-79, Polyquaternium-80, Polyquaternium-81, Polyquaternium-82, Pomaderris Kumerahou Flower/Leaf Extract, Poria *Cocos* Extract, Potassium Abietoyl Hydrolyzed Collagen, Potassium Babassuate, Potassium Behenate, Potassium C9-15 Alkyl Phosphate, Potassium C11-15 Alkyl Phosphate, Potassium C12-13 Alkyl Phosphate, Potassium C12-14 Alkyl Phosphate, Potassium

Caprate, Potassium Capryloyl Glutamate, Potassium Capryloyl Hydrolyzed Rice Protein, Potassium Castorate, Potassium Cocoate, Potassium Cocoyl Glutamate, Potassium Cocoyl Glycinate, Potassium Cocoyl Hydrolyzed Casein, Potassium Cocoyl Hydrolyzed Collagen, Potassium Cocoyl Hydrolyzed Corn Protein, Potassium Cocoyl Hydrolyzed Keratin, Potassium Cocoyl Hydrolyzed Oat Protein, Potassium Cocoyl Hydrolyzed Potato Protein, Potassium Cocoyl Hydrolyzed Rice Bran Protein, Potassium Cocoyl Hydrolyzed Rice Protein, Potassium Cocoyl Hydrolyzed Silk, Potassium Cocoyl Hydrolyzed Soy Protein, Potassium Cocoyl Hydrolyzed Wheat Protein, Potassium Cocoyl Hydrolyzed Yeast Protein, Potassium Cocoyl PCA, Potassium Cocoyl Sarcosinate, Potassium Cocoyl Taurate, Potassium Cornate, Potassium Cyclocarboxypropyloleate, Potassium Dihydroxyethyl Cocamine Oxide Phosphate, Potassium Dimethicone PEG-7 Phosphate, Potassium Dodecylbenzenesulfonate, Potassium Hempseedate, Potassium Hydrogenated Cocoate, Potassium Hydrogenated Palmate, Potassium Hydrogenated Tallowate, Potassium Hydroxystearate, Potassium Isostearate, Potassium Lanolate, Potassium Laurate, Potassium Laureth-3 Carboxylate, Potassium Laureth-4 Carboxylate, Potassium Laureth-5 Carboxylate, Potassium Laureth-6 Carboxylate, Potassium Laureth-10 Carboxylate, Potassium Laureth Phosphate, Potassium Lauroyl Collagen Amino Acids, Potassium Lauroyl Glutamate, Potassium Lauroyl Hydrolyzed Collagen, Potassium Lauroyl Hydrolyzed Pea Protein, Potassium Lauroyl Hydrolyzed Soy Protein, Potassium Lauroyl PCA, Potassium Lauroyl Pea Amino Acids, Potassium Lauroyl Sarcosinate, Potassium Lauroyl Silk Amino Acids, Potassium Lauroyl Wheat Amino Acids, Potassium Lauryl Phosphate, Potassium Lauryl Sulfate, Potassium Linoleate, Potassium Metaphosphate, Potassium Methyl Cocoyl Taurate, Potassium Myristate, Potassium Myristoyl Glutamate, Potassium Myristoyl Hydrolyzed Collagen, Potassium Octoxynol-12 Phosphate, Potassium Oleate, Potassium Oleoyl Hydrolyzed Collagen, Potassium Oliviate, Potassium Olivoyl Hydrolyzed Oat Protein, Potassium Olivoyl Hydrolyzed Wheat Protein, Potassium Olivoyl/Lauroyl Wheat Amino Acids, Potassium Olivoyl PCA, Potassium Palmate, Potassium Palmitate, Potassium Palmitoyl Hydrolyzed Corn Protein, Potassium Palmitoyl Hydrolyzed Oat Protein, Potassium Palmitoyl Hydrolyzed Rice Protein, Potassium Palmitoyl Hydrolyzed Sweet Almond Protein, Potassium Palmitoyl Hydrolyzed Wheat Protein, Potassium Palm Kernelate, Potassium Peanutate, Potassium Rapeseedate, Potassium Ricinoleate, Potassium Safflowerate, Potassium Soyate, Potassium Stearate, Potassium Stearoyl Hydrolyzed Collagen, Potassium Tallate, Potassium Tallowate, Potassium Taurate, Potassium Taurine Laurate, Potassium Trideceth-3 Carboxylate, Potassium Trideceth-4 Carboxylate, Potassium Trideceth-7 Carboxylate, Potassium Trideceth-15 Carboxylate, Potassium Trideceth-19 Carboxylate, Potassium Trideceth-6 Phosphate, Potassium Trideceth-7 Phosphate, Potassium Tsubakiate, Potassium Undecylenate, Potassium Undecylenoyl Hydrolyzed Collagen, Potassium Undecylenoyl Hydrolyzed Rice Protein, PPG-30-Buteth-30, PPG-36-Buteth-36, PPG-38-Buteth-37, PPG-30-Capryleth-4 Phosphate, PPG-10 Cetyl Ether Phosphate, PPG-2 C9-11 Pareth-8, PPG-1-Deceth-5, PPG-3-Deceth-2 Carboxylic Acid, PPG-30 Ethylhexeth-4 Phosphate, PPG-20-Glycereth-30, PPG-2 Hydroxyethyl Coco/Isostearamide, PPG-2-Isodeceth-8, PPG-2-Isodeceth-10, PPG-2-Isodeceth-18, PPG-2-Isodeceth-25, PPG-4-Isodeceth-10, Propyltrimonium Hydrolyzed Collagen, Quaternium-24, Quaternium-52, Quaternium-87, Rapeseed Acid, Rice Bran Acid, Rice Oil

Glycereth-8 Esters, Ricinoleamidopropyl Betaine, Ricinoleic Acid, Ricinoleth-40, Safflower Acid, Sapindus *Oahuensis* Fruit Extract, *Saponaria Officinalis* Root Powder, Saponins, Sekken-K, Sekken-Na/K, Sekken Soji, Sekken SojiK, Sesame Oil Glycereth-8 Esters, Sesamidopropylamine Oxide, Sesamidopropyl Betaine, Shea Butteramidopropyl Betaine, Shea Butter Glycereth-8 Esters, Sodium Arachidate, Sodium Arganampohoacetate, Sodium Astrocaryum Murumurate, Sodium Avocadoate, Sodium Babassuamphoacetate, Sodium Babassuate, Sodium Babassu Sulfate, Sodium Behenate, Sodium Bisglycol Ricinosulfosuccinate, Sodium Bis-Hydroxyethylglycinate Coco-Glucosides Crosspolymer, Sodium Bis-Hydroxyethylglycinate Lauryl-Glucosides Crosspolymer, Sodium Borageamidopropyl PG-Dimonium Chloride Phosphate, Sodium Butoxynol-12 Sulfate, Sodium Butylglucosides Hydroxypropyl Phosphate, Sodium C13-17 Alkane Sulfonate, Sodium C14-18 Alkane Sulfonate, Sodium C12-15 Alkoxypropyl Iminodipropionate, Sodium C10-16 Alkyl Sulfate, Sodium C11-15 Alkyl Sulfate, Sodium C12-13 Alkyl Sulfate, Sodium C12-15 Alkyl Sulfate, Sodium C12-18 Alkyl Sulfate, Sodium C16-20 Alkyl Sulfate, Sodium C9-22 Alkyl Sec Sulfonate, Sodium C14-17 Alkyl Sec Sulfonate, Sodium Caprate, Sodium Caproamphoacetate, Sodium Caproamphohydroxypropylsulfonate, Sodium Caproamphopropionate, Sodium Caproyl Methyltaurate, Sodium Caprylate, Sodium Capryleth-2 Carboxylate, Sodium Capryleth-9 Carboxylate, Sodium Capryloamphoacetate, Sodium Capryloamphohydroxypropylsulfonate, Sodium Capryloamphopropionate, Sodium Capryloyl Glutamate, Sodium Capryloyl Hydrolyzed Wheat Protein, Sodium Caprylyl PG-Sulfonate, Sodium Caprylyl Sulfonate, Sodium Castorate, Sodium Ceteareth-13 Carboxylate, Sodium Cetearyl Sulfate, Sodium Ceteth-13 Carboxylate, Sodium Cetyl Sulfate, Sodium Cocamidopropyl PG-Dimonium Chloride Phosphate, Sodium Cocaminopropionate, Sodium Coceth Sulfate, Sodium Coceth-30 Sulfate, Sodium Cocoabutteramphoacetate, Sodium Cocoa Butterate, Sodium Cocoamphoacetate, Sodium Cocoamphohydroxypropylsulfonate, Sodium Cocoamphopropionate, Sodium Cococate, Sodium Coco/Babassu/Andiroba Sulfate, Sodium Coco/Babassu Sulfate, Sodium Cocoglucosides Hydroxypropyl Phosphate, Sodium Cocoglucosides Hydroxypropylsulfonate, Sodium Coco-Glucoside Tartrate, Sodium Cocoglyceryl Ether Sulfonate, Sodium Coco/Hydrogenated Tallow Sulfate, Sodium Cocoiminodiacetate, Sodium Cocomonoglyceride Sulfate, Sodium Cocomonoglyceride Sulfonate, Sodium Coco PG-Dimonium Chloride Phosphate, Sodium Coco-Sulfate, Sodium Coco Sulfoacetate, Sodium Cocoyl Alaninate, Sodium Cocoyl Amino Acids, Sodium Cocoyl Collagen Amino Acids, Sodium Cocoyl Glutamate, Sodium Cocoyl Glutamate, Sodium Cocoyl Glycinate, Sodium Cocoyl/Hydrogenated Tallow Glutamate, Sodium Cocoyl Hydrolyzed Collagen, Sodium Cocoyl Hydrolyzed Keratin, Sodium Cocoyl Hydrolyzed Rice Protein, Sodium Cocoyl Hydrolyzed Silk, Sodium Cocoyl Hydrolyzed Soy Protein, Sodium Cocoyl Hydrolyzed Sweet Almond Protein, Sodium Cocoyl Hydrolyzed Wheat Protein, Sodium Cocoyl Hydrolyzed Wheat Protein Glutamate, Sodium Cocoyl Isethionate, Sodium Cocoyl Methylaminopropionate, Sodium Cocoyl Oat Amino Acids, Sodium Cocoyl/Palmoyl/Sunfloweroyl Glutamate, Sodium Cocoyl Proline, Sodium Cocoyl Sarcosinate, Sodium Cocoyl Taurate, Sodium Cocoyl Threoninate, Sodium Cocoyl Wheat Amino Acids, Sodium C12-14 Olefin Sulfonate, Sodium C14-16 Olefin Sulfonate, Sodium C14-18 Olefin Sulfonate, Sodium C16-18 Olefin Sulfonate, Sodium Cornamphopropionate, Sodium Cottonseed-

amphoacetate, Sodium C13-15 Pareth-8 Butyl Phosphate, Sodium C9-11 Pareth-6 Carboxylate, Sodium C11-15 Pareth-7 Carboxylate, Sodium C12-13 Pareth-5 Carboxylate, Sodium C12-13 Pareth-8 Carboxylate, Sodium C12-13 Pareth-12 Carboxylate, Sodium C12-15 Pareth-6 Carboxylate, Sodium C12-15 Pareth-7 Carboxylate, Sodium C12-15 Pareth-8 Carboxylate, Sodium C14-15 Pareth-8 Carboxylate, Sodium C12-14 Sec-Pareth-8 Carboxylate, Sodium C14-15 Pareth-PG Sulfonate, Sodium C12-13 Pareth-2 Phosphate, Sodium C13-15 Pareth-8 Phosphate, Sodium C9-15 Pareth-3 Sulfate, Sodium C10-15 Pareth Sulfate, Sodium C10-16 Pareth-2 Sulfate, Sodium C12-13 Pareth Sulfate, Sodium C12-15 Pareth Sulfate, Sodium C12-15 Pareth-3 Sulfate, Sodium C13-15 Pareth-3 Sulfate, Sodium C12-14 Sec-Pareth-3 Sulfate, Sodium C12-15 Pareth-3 Sulfonate, Sodium C12-15 Pareth-7 Sulfonate, Sodium C12-15 Pareth-15 Sulfonate, Sodium Deceth-2 Carboxylate, Sodium Deceth Sulfate, Sodium Decylbenzenesulfonate, Sodium Decylglucosides Hydroxypropyl Phosphate, Sodium Decylglucosides Hydroxypropylsulfonate, Sodium Dilaureth-7 Citrate, Sodium Dilaureth-10 Phosphate, Sodium Dilinoleamidopropyl PG-Dimonium Chloride Phosphate, Sodium Dilinoleate, Sodium Dioleth-8 Phosphate, Sodium Dodecylbenzenesulfonate, Sodium Ethyl 2-Sulfolaurate, Sodium Glyceryl Oleate Phosphate, Sodium Grapeseedamidopropyl PG-Dimonium Chloride Phosphate, Sodium Grapeseedamphoacetate, Sodium Grapeseedate, Sodium Hempseedamphoacetate, Sodium Hexeth-4 Carboxylate, Sodium Hydrogenated Cococate, Sodium Hydrogenated Cocoyl Methyl Isethionate, Sodium Hydrogenated Palmate, Sodium Hydrogenated Tallowate, Sodium Hydrogenated Tallowoyl Glutamate, Sodium Hydroxylauryl-dimonium Ethyl Phosphate, Sodium Hydroxypropyl Palm Kernelate Sulfonate, Sodium Hydroxypropylphosphate Decylglucoside Crosspolymer, Sodium Hydroxypropylphosphate Laurylglucoside Crosspolymer, Sodium Hydroxypropylsulfonate Cocoglucoside Crosspolymer, Sodium Hydroxypropylsulfonate Decylglucoside Crosspolymer, Sodium Hydroxypropylsulfonate Laurylglucoside Crosspolymer, Sodium Hydroxystearate, Sodium Isostearate, Sodium Isosteareth-6 Carboxylate, Sodium Isosteareth-11 Carboxylate, Sodium Isostearoamphoacetate, Sodium Isostearoamphopropionate, Sodium N-Isostearoyl Methyltaurate, Sodium Laneth Sulfate, Sodium Lanolate, Sodium Lardate, Sodium Lauramido Diacetate, Sodium Lauraminopropionate, Sodium Laurate, Sodium Laureth-3 Carboxylate, Sodium Laureth-4 Carboxylate, Sodium Laureth-5 Carboxylate, Sodium Laureth-6 Carboxylate, Sodium Laureth-8 Carboxylate, Sodium Laureth-11 Carboxylate, Sodium Laureth-12 Carboxylate, Sodium Laureth-13 Carboxylate, Sodium Laureth-14 Carboxylate, Sodium Laureth-16 Carboxylate, Sodium Laureth-17 Carboxylate, Sodium Laureth Sulfate, Sodium Laureth-5 Sulfate, Sodium Laureth-7 Sulfate, Sodium Laureth-8 Sulfate, Sodium Laureth-12 Sulfate, Sodium Laureth-40 Sulfate, Sodium Laureth-7 Tartrate, Sodium Lauriminodipropionate, Sodium Lauroamphoacetate, Sodium Lauroamphohydroxypropylsulfonate, Sodium Lauroampho PG-Acetate Phosphate, Sodium Lauroamphopropionate, Sodium Lauroyl Aspartate, Sodium Lauroyl Collagen Amino Acids, Sodium Lauroyl Glycine Propionate, Sodium Lauroyl Hydrolyzed Collagen, Sodium Lauroyl Hydrolyzed Silk, Sodium Lauroyl Hydroxypropyl Sulfonate, Sodium Lauroyl Isethionate, Sodium Lauroyl Methylaminopropionate, Sodium Lauroyl Methyl Isethionate, Sodium Lauroyl Millet Amino Acids, Sodium Lauroyl/Myristoyl Aspartate, Sodium Lauroyl Oat Amino Acids, Sodium Lauroyl Sarcosinate, Sodium Lauroyl Silk Amino

Acids, Sodium Lauroyl Taurate, Sodium Lauroyl Wheat Amino Acids, Sodium Lauryl Diethylenediaminoglycinate, Sodium Lauryl Glucose Carboxylate, Sodium Laurylglucosides Hydroxypropyl Phosphate, Sodium Laurylglucosides Hydroxypropylsulfonate, Sodium Lauryl Glycol Carboxylate, Sodium Lauryl Hydroxyacetamide Sulfate, Sodium Lauryl Phosphate, Sodium Lauryl Sulfate, Sodium Lauryl Sulfoacetate, Sodium Linoleate, Sodium Macadamiasseedate, Sodium Mangoamphoacetate, Sodium Mangoseedate, Sodium/MEA Laureth-2 Sulfosuccinate, Sodium Methoxy PPG-2 Acetate, Sodium Methyl Cocoyl Taurate, Sodium Methyl Lauroyl Taurate, Sodium Methyl Myristoyl Taurate, Sodium Methyl Oleoyl Taurate, Sodium Methyl Palmitoyl Taurate, Sodium Methyl Stearoyl Taurate, Sodium Methyl 2-Sulfolaurate, Sodium Methyl 2-Sulfopalmitate, Sodium Methyltaurate Isopalmitamide, Sodium Methyltaurine Cocoyl Methyltaurate, Sodium Myreth Sulfate, Sodium Myristate, Sodium Myristoamphoacetate, Sodium Myristoyl Glutamate, Sodium Myristoyl Hydrolyzed Collagen, Sodium Myristoyl Isethionate, Sodium Myristoyl Sarcosinate, Sodium Myristyl Sulfate, Sodium Nonoxynol-6 Phosphate, Sodium Nonoxynol-9 Phosphate, Sodium Nonoxynol-1 Sulfate, Sodium Nonoxynol-3 Sulfate, Sodium Nonoxynol-4 Sulfate, Sodium Nonoxynol-6 Sulfate, Sodium Nonoxynol-8 Sulfate, Sodium Nonoxynol-10 Sulfate, Sodium Nonoxynol-25 Sulfate, Sodium Octoxynol-2 Ethane Sulfonate, Sodium Octoxynol-2 Sulfate, Sodium Octoxynol-6 Sulfate, Sodium Octoxynol-9 Sulfate, Sodium Oleate, Sodium Oleoamphoacetate, Sodium Oleoamphohydroxypropylsulfonate, Sodium Oleoamphopropionate, Sodium Oleoyl Hydrolyzed Collagen, Sodium Oleoyl Isethionate, Sodium Oleth Sulfate, Sodium Oleyl Methyl Isethionate, Sodium Oleyl Sulfate, Sodium Olivamphoacetate, Sodium Olivate, Sodium Olivoyl Glutamate, Sodium Palmamphoacetate, Sodium Palmate, Sodium Palm Glyceride Sulfonate, Sodium Palmitate, Sodium Palmitoyl Hydrolyzed Collagen, Sodium Palmitoyl Hydrolyzed Wheat Protein, Sodium Palmitoyl Sarcosinate, Sodium Palm Kernelate, Sodium Palm Kerneloyl Isethionate, Sodium Palmoyl Glutamate, Sodium *Passiflora Edulis* Seedate, Sodium Peanutamphoacetate, Sodium Peanutate, Sodium PEG-6 Cocamide Carboxylate, Sodium PEG-8 Cocamide Carboxylate, Sodium PEG-4 Cocamide Sulfate, Sodium PEG-3 Lauramide Carboxylate, Sodium PEG-4 Lauramide Carboxylate, Sodium PEG-8 Palm Glycerides Carboxylate, Sodium Pentaerythrityl Hydroxypropyl Iminodiacetate Dendrimer, Sodium Propoxy PPG-2 Acetate, Sodium Rapeseedate, Sodium Ricebranamphoacetate, Sodium Ricinoleate, Sodium Ricinoleoamphoacetate, Sodium Rose Hipsamphoacetate, Sodium Rosinate, Sodium Safflowerate, Sodium Saffloweroyl Hydrolyzed Soy Protein, Sodium Sesameseedate, Sodium Sesamphoacetate, Sodium Sheabutteramphoacetate, Sodium Soyate, Sodium Soy Hydrolyzed Collagen, Sodium Stearate, Sodium Stearoamphoacetate, Sodium Stearoamphohydroxypropylsulfonate, Sodium Stearoamphopropionate, Sodium Stearoyl Casein, Sodium Stearoyl Glutamate, Sodium Stearoyl Hyaluronate, Sodium Stearoyl Hydrolyzed Collagen, Sodium Stearoyl Hydrolyzed Corn Protein, Sodium Stearoyl Hydrolyzed Silk, Sodium Stearoyl Hydrolyzed Soy Protein, Sodium Stearoyl Hydrolyzed Wheat Protein, Sodium Stearoyl Lactalbumin, Sodium Stearoyl Methyl Isethionate, Sodium Stearoyl Oat Protein, Sodium Stearoyl Pea Protein, Sodium Stearoyl Soy Protein, Sodium Stearyl Dimethyl Glycine, Sodium Stearyl Sulfate, Sodium Sunflowerseedamphoacetate, Sodium Surfactin, Sodium Sweetalmondamphoacetate, Sodium Sweet Almondate, Sodium Tallamphopropionate, Sodium Tallate, Sodium

Tallowamphoacetate, Sodium Tallowate, Sodium Tallow Sulfate, Sodium Tamanuseedate, Sodium Taurate, Sodium Taurine Cocoyl Methyltaurate, Sodium Taurine Laurate, Sodium/TEA-Lauroyl Collagen Amino Acids, Sodium/TEA-Lauroyl Hydrolyzed Collagen, Sodium/TEA-Lauroyl Hydrolyzed Keratin, Sodium/TEA-Lauroyl Keratin Amino Acids, Sodium/TEA-Undecylenoyl Collagen Amino Acids, Sodium/TEA-Undecylenoyl Hydrolyzed Collagen, Sodium/TEA-Undecylenoyl Hydrolyzed Corn Protein, Sodium/TEA-Undecylenoyl Hydrolyzed Soy Protein, Sodium/TEA-Undecylenoyl Hydrolyzed Wheat Protein, Sodium *Theobroma Grandiflorum* Seedate, Sodium Trideceth-3 Carboxylate, Sodium Trideceth-4 Carboxylate, Sodium Trideceth-6 Carboxylate, Sodium Trideceth-7 Carboxylate, Sodium Trideceth-8 Carboxylate, Sodium Trideceth-12 Carboxylate, Sodium Trideceth-15 Carboxylate, Sodium Trideceth-19 Carboxylate, Sodium Trideceth Sulfate, Sodium Tridecylbenzenesulfonate, Sodium Tridecyl Sulfate, Sodium Trimethylolpropane Hydroxypropyl Iminodiacetate Dendrimer, Sodium Undeceth-5 Carboxylate, Sodium Undecylenate, Sodium Undecylenoamphoacetate, Sodium Undecylenoamphopropionate, Sodium Undecylenoyl Glutamate, Sodium Wheat Germamphoacetate, Sorbeth-160 Tristearate, SoyAcid, Soyamidopropylamine Oxide, Soyamidopropyl Betaine, Soybean Oil Glycereth-8 Esters, Stearamidopropylamine Oxide, Stearamidopropyl Betaine, Stearamine Oxide, Steareth-15, Steareth-16, Steareth-20, Steareth-21, Steareth-25, Steareth-27, Steareth-30, Steareth-40, Steareth-50, Steareth-80, Steareth-100, Steareth-2 Phosphate, Steareth-3 Phosphate, Stearic Acid, Stearoxypolytrimonium Chloride, Stearoyl Glutamic Acid, Stearoyl Sarcosine, Stearyl Betaine, Stearyldimoniumhydroxypropyl Butylglucosides Chloride, Stearyldimoniumhydroxypropyl Decylglucosides Chloride, Stearyldimoniumhydroxypropyl Laurylglucosides Chloride, Sulfated Castor Oil, Sulfated Coconut Oil, Sulfated Glyceryl Oleate, Sulfated Olive Oil, Sulfated Peanut Oil, Sunfloweramide MEA, SunflowerSeed Acid, Sunflowerseedamidopropyl Hydroxyethyltrimonium Chloride, Sunflower Seed Oil Glycereth-8 Esters, Tall Oil Acid, Tallow Acid, Tallowamidopropylamine Oxide, Tallowamidopropyl Betaine, Tallowamidopropyl Hydroxysulfate, Tallowamine Oxide, Tallow Betaine, Tallow Dihydroxyethyl Betaine, Tallowoyl Ethyl Glucoside, TEA-Abietoyl Hydrolyzed Collagen, TEA-C12-14 Alkyl Phosphate, TEA-C10-15 Alkyl Sulfate, TEA-C11-15 Alkyl Sulfate, TEA-C12-13 Alkyl Sulfate, TEA-C12-14 Alkyl Sulfate, TEA-C12-15 Alkyl Sulfate, TEA C14-17 Alkyl Sec Sulfonate, TEA-Canolate, TEACocamide Diacetate, TEA-Cococate, TEA-Coco-Sulfate, TEA-Cocoyl Alaninate, TEA-Cocoyl Glutamate, TEA-Cocoyl Glutamate, TEA-Cocoyl Glycinate, TEA-Cocoyl Hydrolyzed Collagen, TEACocoyl Hydrolyzed Soy Protein, TEA-Cocoyl Sarcosinate, TEA-Dimethicone PEG-7 Phosphate, TEA-Dodecylbenzenesulfonate, TEA-Hydrogenated Cococate, TEA-Hydrogenated Tallowoyl Glutamate, TEA-Isostearate, TEA-Isostearoyl Hydrolyzed Collagen, TEA-Lauraminopropionate, TEA-Laurate, TEA-Laurate/Myristate, TEA-Laureth Sulfate, TEA-Lauroyl Collagen Amino Acids, TEA-Lauroyl Glutamate, TEA-Lauroyl Hydrolyzed Collagen, TEA-Lauroyl Keratin Amino Acids, TEA-Lauroyl Methylaminopropionate, TEA-Lauroyl/Myristoyl Aspartate, TEA-Lauroyl Sarcosinate, TEA-Lauryl Phosphate, TEA-Lauryl Sulfate, TEA-Myristaminopropionate, TEA-Myristate, TEA-Myristoyl Hydrolyzed Collagen, TEA-Oleate, TEA-Oleoyl Hydrolyzed Collagen, TEA-Oleoyl Sarcosinate, TEA-Oleoyl Sulfate, TEA-Palmitate, TEA-Palm Kernel Sarcosinate, TEAPEG-3 Cocamide Sulfate, TEA-Rosinate, TEA-Stear-

ate, TEA-Tallate, TEA-T ridecylbenzenesulfonate, TEA-Undecylenate, TEA-Undecylenoyl Hydrolyzed Collagen, Tetramethyl Decynediol, Tetrasodium Dicarboxyethyl Stearyl Sulfosuccinamate, TIPA-Laureth Sulfate, TIPA-Lauryl Sulfate, TIPA-Myristate, TIPA-Stearate, Tocopheryl Phosphate, Trehalose Undecylenoate, TMC12-15 Pareth-2 Phosphate, TM-C12-15 Pareth-6 Phosphate, TM-C12-15 Pareth-8 Phosphate, TM-C12-15 Pareth-10 Phosphate, Trideceth-20, Trideceth-50, Trideceth-3 Carboxylic Acid, Trideceth-4 Carboxylic Acid, Trideceth-7 Carboxylic Acid, Trideceth-8 Carboxylic Acid, Trideceth-15 Carboxylic Acid, Trideceth-19 Carboxylic Acid, Trideceth-10 Phosphate, Tridecylbenzenesulfonic Acid, Trilaureth-9 Citrate, Trimethylolpropane Hydroxypropyl Bis-Hydroxyethylamine Dendrimer, Trisodium Lauroampho PG-Acetate Chloride Phosphate, Undecanoic Acid, Undeceth-5 Carboxylic Acid, Undecylenamidopropylamine Oxide, Undecylenamidopropyl Betaine, Undecylenic Acid, Undecylenoyl Collagen Amino Acids, Undecylenoyl Glycine, Undecylenoyl Hydrolyzed Collagen, Undecylenoyl Wheat Amino Acids, Undecyl Glucoside, Wheat Germ Acid, Wheat Germamidopropylamine Oxide, Wheat Germamidopropyl Betaine, *Yucca Schidigera* Leaf/Root/Stem Extract, *Yucca Schidigera* Stem Extract, Zinc Coceth Sulfate and Zinc Coco-Sulfate. Non-Ionic (Co-)Surfactants

Alcohol alkoxylates. The added nonionic surfactants are preferably alkoxylated and/or propoxylated, particularly primary alcohols having preferably 8 to 18 carbon atoms and an average of 1 to 12 mol ethylene oxide (EO) and/or 1 to 10 mol propylene oxide (PO) per mol alcohol. C₈-C₁₆-Alcohol alkoxylates, advantageously ethoxylated and/or propoxylated C₁₀-C₁₅-alcohol alkoxylates, particularly C₁₂-C₁₄ alcohol alkoxylates, with an ethoxylation degree between 2 and 10, preferably between 3 and 8, and/or a propoxylation degree between 1 and 6, preferably between 1.5 and 5, are particularly preferred. The cited degrees of ethoxylation and propoxylation constitute statistical average values that can be a whole or a fractional number for a specific product. Preferred alcohol ethoxylates and propoxylates have a narrowed homolog distribution (narrow range ethoxylates/propoxylates, NRE/NRP). In addition to these nonionic surfactants, fatty alcohols with more than 12 EO can also be used. Examples of these are (tallow) fatty alcohols with 14 EO, 16 EO, 20 EO, 25 EO, 30 EO or 40 EO.

Alkylglycosides (APG®). Furthermore, as additional nonionic surfactants, alkyl glycosides that satisfy the general Formula RO(G)_x, can be added, e.g., as compounds, particularly with anionic surfactants, in which R means a primary linear or methyl-branched, particularly 2-methyl-branched, aliphatic group containing 8 to 22, preferably 12 to 18 carbon atoms and G stands for a glucose unit containing 5 or 6 carbon atoms, preferably for glucose. The degree of oligomerization x, which defines the distribution of monoglycosides and oligoglycosides, is any number between 1 and 10, preferably between 1.1 and 1.4.

Fatty acid ester alkoxylates. Another class of preferred nonionic surfactants, which are used either as the sole nonionic surfactant or in combination with other nonionic surfactants, in particular, together with alkoxylated fatty alcohols and/or alkyl glycosides, are alkoxylated, preferably ethoxylated or ethoxylated and propoxylated fatty acid alkyl esters preferably containing 1 to 4 carbon atoms in the alkyl chain, more particularly the fatty acid methyl esters which are described, for example, in Japanese Patent Application JP-A58/217598 or which are preferably produced by the process described in International Patent Application WO-A-90/13533. Methyl esters of C₁₂-C₁₈ fatty acids con-

taining an average of 3 to EO, particularly containing an average of 5 to 12 EO, are particularly preferred.

Amine oxides. Nonionic surfactants of the amine oxide type, for example, N-coco alkyl-N,N-dimethylamine oxide and N-tallow alkyl-N,N-dihydroxyethylamine oxide, and the fatty acid alkanolamides may also be suitable. The quantity in which these nonionic surfactants are used is preferably no more than the quantity in which the ethoxylated fatty alcohols are used and, particularly no more than half that quantity.

Gemini surfactants. The so-called gemini surfactants can be considered as further surfactants. Generally speaking, such compounds are understood to mean compounds that have two hydrophilic groups and two hydrophobic groups per molecule. As a rule, these groups are separated from one another by a "spacer". The spacer is usually a hydrocarbon chain that is intended to be long enough such that the hydrophilic groups are a sufficient distance apart to be able to act independently of one another. These types of surfactants are generally characterized by an unusually low critical micelle concentration and the ability to strongly reduce the surface tension of water. In exceptional cases, however, not only dimeric but also trimeric surfactants are meant by the term gemini surfactants. Suitable gemini surfactants are, for example, sulfated hydroxy mixed ethers according to German Patent Application DE 4321022 A1 or dimer alcohol bis- and trimer alcohol trisulfates and ether sulfates according to International Patent Application WO 96/23768 A1. Blocked end group dimeric and trimeric mixed ethers according to German Patent Application DE 19513391 A1 are especially characterized by their bifunctionality and multifunctionality. Gemini polyhydroxy fatty acid amides or polyhydroxy fatty acid amides, such as those described in International Patent Applications WO 95/19953 A1, WO 95/19954 A1 and WO 95/19955 A1 can also be used.

Cationic Co-Surfactants

Tetraalkyl ammonium salts. Cationically active surfactants comprise the hydrophobic high molecular group required for the surface activity in the cation by dissociation in aqueous solution. A group of important representatives of the cationic surfactants are the tetraalkyl ammonium salts of the general formula: (R¹R²R³R⁴N⁺)X⁻. Here R¹ stands for C₁-C₈ alk(en)yl, R², R³ and R⁴, independently of each other, for alk(en)yl radicals having 1 to 22 carbon atoms. X is a counter ion, preferably selected from the group of the halides, alkyl sulfates and alkyl carbonates. Cationic surfactants, in which the nitrogen group is substituted with two long acyl groups and two short alk(en)yl groups, are particularly preferred.

Esterquats. A further class of cationic surfactants particularly useful as co-surfactants for the present invention is represented by the so-called esterquats. Esterquats are generally understood to be quaternised fatty acid triethanolamine ester salts. These are known compounds which can be obtained by the relevant methods of preparative organic chemistry. Reference is made in this connection to International patent application WO 91/01295 A1, according to which triethanolamine is partly esterified with fatty acids in the presence of hypophosphorous acid, air is passed through the reaction mixture and the whole is then quaternised with dimethyl sulphate orethylene oxide. In addition, German patent DE 4308794 C1 describes a process for the production of solid esterquats in which the quaternisation of triethanolamine esters is carried out in the presence of suitable dispersants, preferably fatty alcohols.

Typical examples of esterquats suitable for use in accordance with the invention are products of which the acyl

component derives from monocarboxylic acids corresponding to formula RCOOH in which RCO is an acyl group containing 6 to 10 carbon atoms, and the amine component is triethanolamine (TEA). Examples of such monocarboxylic acids are caproic acid, caprylic acid, capric acid and technical mixtures thereof such as, for example, so-called head-fractionated fatty acid. Esterquats of which the acyl component derives from monocarboxylic acids containing 8 to 10 carbon atoms, are preferably used. Other esterquats are those of which the acyl component derives from dicarboxylic acids like malonic acid, succinic acid, maleic acid, fumaric acid, glutaric acid, sorbic acid, pimelic acid, azelaic acid, sebacic acid and/or dodecanedioic acid, but preferably adipic acid. Overall, esterquats of which the acyl component derives from mixtures of monocarboxylic acids containing 6 to 22 carbon atoms, and adipic acid are preferably used. The molar ratio of mono and dicarboxylic acids in the final esterquat may be in the range from 1:99 to 99:1 and is preferably in the range from 50:50 to 90:10 and more particularly in the range from 70:30 to 80:20. Besides the quaternised fatty acid triethanolamine ester salts, other suitable esterquats are quaternised ester salts of mono/dicarboxylic acid mixtures with diethanolalkylamines or 1,2-dihydroxypropyl dialkylamines. The esterquats may be obtained both from fatty acids and from the corresponding triglycerides in admixture with the corresponding dicarboxylic acids. One such process, which is intended to be representative of the relevant prior art, is proposed in European patent EP 0750606 B1. To produce the quaternised esters, the mixtures of mono- and dicarboxylic acids and the triethanolamine—based on the available carboxyl functions—may be used in a molar ratio of 1.1:1 to 3:1. With the performance properties of the esterquats in mind, a ratio of 1.2:1 to 2.2:1 and preferably 1.5:1 to 1.9:1 has proved to be particularly advantageous. The preferred esterquats are technical mixtures of mono-, di- and triesters with an average degree of esterification of 1.5 to 1.9.

Amphoteric Co-Surfactants

Betaines. Amphoteric or ampholytic surfactants possess a plurality of functional groups that can ionize in aqueous solution and thereby—depending on the conditions of the medium—lend anionic or cationic character to the compounds (see DIN 53900, July 1972). Close to the isoelectric point (around pH 4), the amphoteric surfactants form inner salts, thus becoming poorly soluble or insoluble in water. Amphoteric surfactants are subdivided into ampholytes and betaines, the latter existing as zwitterions in solution. Ampholytes are amphoteric electrolytes, i.e. compounds that possess both acidic as well as basic hydrophilic groups and therefore behave as acids or as bases depending on the conditions. Especially betaines are known surfactants which are mainly produced by carboxyalkylation, preferably carboxymethylation, of amine compounds. The starting materials are preferably condensed with halocarboxylic acids or salts thereof, more particularly sodium chloroacetate, one mole of salt being formed per mole of betaine. The addition of unsaturated carboxylic acids, such as acrylic acid for example, is also possible. Examples of suitable betaines are the carboxy alkylation products of secondary and, in particular, tertiary amines which correspond to formula $\text{R}^1\text{R}^2\text{R}^3\text{N}-(\text{CH}_2)_q\text{COOX}$ where R^1 is an alkyl radical having 6 to 22 carbon atoms, R^2 is hydrogen or an alkyl group containing 1 to 4 carbon atoms, R^3 is an alkyl group containing 1 to 4 carbon atoms, q is a number of 1 to 6 and X is an alkali and/or alkaline earth metal or ammonium. Typical examples are the carboxymethylation products of hexylmethylethylamine, hexyldimethylamine, octyldimethylam-

ine, decyldimethylamine, $\text{C}_{12/14}$ -cocoalkyldimethylamine, myristyldimethylamine, cetyldimethylamine, stearyldimethylamine, stearylethylmethylamine, oleyldimethylamine, $\text{C}_{17/19}$ -tallowalkyldimethylamine and their technical mixtures, and particularly dodecyl methylamine, dodecyl dimethylamine, dodecyl ethylmethylamine and technical mixtures thereof.

Alkylamido betaines. Other suitable betaines are the carboxyalkylation products of amidoamines corresponding to formula $\text{R}^1\text{CO}(\text{R}^2)(\text{R}^3)-\text{NH}-(\text{CH}_2)_p-\text{N}-(\text{CH}_2)_q\text{COOX}$ in which R^1CO is an aliphatic acyl radical having 6 to 22 carbon atoms and 0 or 1 to 3 double bonds, R^2 is hydrogen or an alkyl radical having 1 to 4 carbon atoms, R^3 is an alkyl radical having 1 to 4 carbon atoms, p is a number from 1 to 6, q is a number from 1 to 3 and X is an alkali and/or alkaline earth metal or ammonium. Typical examples are reaction products of fatty acids having 6 to 22 carbon atoms, like for example caproic acid, caprylic acid, caprinic acid, lauric acid, myristic acid, palmitic acid, palmoleic acid, stearic acid, isostearic acid, oleic acid, elaidic acid, petroselinic acid, linolic acid linoleic acid, elaeostearic acid, arachidonic acid, gadoleic acid, behenic acid, erucic acid and their technical mixtures with N,N -dimethylaminoethylamine, N,N -dimethylaminopropylamine, N,N -diethylaminoethylamine und N,N -diethylaminopropylamine, which are condensed with sodium chloroacetate. The commercially available products include Dehyton® K and Dehyton® PK (Cognis Deutschland GmbH & Co., KG) as well as Tego® Betaine (Goldschmidt).

Imidazolines. Other suitable starting materials for the betaines to be used for the purposes of the invention are imidazolines. These substances are also known and may be obtained, for example, by cyclizing condensation of 1 or 2 moles of C_6 - C_{22} fatty acids with polyfunctional amines, such as for example aminoethyl ethanolamine (AEEA) or diethylenetriamine. The corresponding carboxyalkylation products are mixtures of different open-chain betaines. Typical examples are condensation products of the above-mentioned fatty acids with AEEA, preferably imidazolines based on lauric acid, which are subsequently betainised with sodium chloroacetate. The commercially available products include Dehyton® G (Cognis Deutschland GmbH & Co., KG).

The amount of (co-)surfactant comprised in the inventive compositions is advantageously 0.1 wt. % to 90 wt. %, particularly 10 wt. % to 80 wt. % and particularly preferably 20 wt. % to 70 wt.-%.

Organic Solvents

Liquid light or heavy duty detergents may comprise organic solvents, preferably those miscible with water. Polydiols, ethers, alcohols, ketones, amides and/or esters are preferably used as the organic solvent for this in amounts of 0 to 90 wt. %, preferably 0.1 to 70 wt. %, particularly 0.1 to 60 wt. %. Low molecular weight polar substances, such as for example, methanol, ethanol, propylene carbonate, acetone, acetonylacetone, diacetone alcohol, ethyl acetate, 2-propanol, ethylene glycol, propylene glycol, glycerin, diethylene glycol, dipropylene glycol monomethyl ether and dimethylformamide or their mixtures are preferred.

Enzymes

Cellulase Enzymes. Cellulase enzymes optionally used in the instant detergent composition are preferably incorporated, when present, at levels sufficient to provide up to about 5 mg by weight, more preferably about 0.01 mg to about 3 mg, of active enzyme per gram of the composition. Unless stated otherwise, the compositions herein preferably

comprise from about 0.001% to about 5%, preferably 0.01%-1% by weight of a commercial enzyme preparation.

The cellulases suitable for the present invention include either bacterial or fungal cellulase. Preferably, they will have a pH optimum of between 5 and 9.5. Suitable cellulases are fungal cellulase produced from *Humicola insolens* and *Humicola* strain DSM1800 or a cellulase 212-producing fungus belonging to the genus *Aeromonas*, and cellulase extracted from the hepatopancreas of a marine mollusk (*Dolabella Auricula Solander*), suitable cellulases are also disclosed in GB 2,075,028A. In addition, cellulase especially suitable for use herein are disclosed in WO 1992013057A1. Most preferably, the cellulases used in the instant detergent compositions are purchased commercially from NOVO Industries A/S under the product names CAREZYMEO and CELLUZYMEO.

Other Enzymes. Additional enzymes can be included in the detergent compositions herein for a wide variety of fabric laundering purposes, including removal of protein-based, carbohydrate-based, or triglyceride-based stains, for example, and for the prevention of refugee dye transfer, and for fabric restoration. The additional enzymes to be incorporated include proteases, amylases, lipases, and peroxidases, as well as mixtures thereof. Other types of enzymes can also be included. They can be of any suitable origin, such as vegetable, animal, bacterial, fungal and yeast origin. However, their choice is governed by several factors such as pH-activity and/or stability optima, thermostability, stability versus active detergents, builders as well as their potential to cause malodors during use. In this respect bacterial or fungal enzymes are preferred, such as bacterial amylases and proteases.

Enzymes are normally incorporated at levels sufficient to provide up to about 5 mg by weight, more typically about 0.01 mg to about 3 mg, of active enzyme per gram of the composition. Stated otherwise, the compositions herein will typically comprise from about 0.001% to about 5%, preferably 0.01%-1% by weight of a commercial enzyme preparation. Protease enzymes are usually present in such commercial preparations at levels sufficient to provide from 0.005 to 0.1 Anson units (AU) of activity per gram of composition.

Suitable examples of proteases are the subtilisins which are obtained from particular strains of *B. subtilis* and *B. licheniformis*. Another suitable protease is obtained from a strain of *Bacillus*, having maximum activity throughout the pH range of 8-12, developed and sold by Novo Industries A/S under the registered trade name ESPERASE®. The preparation of this enzyme and analogous enzymes is described in GB 1,243,784 of Novo. Proteolytic enzymes suitable for removing protein-based stains that are commercially available include those sold under the trade names ALCALASE® and SAVINASE® by Novo Industries A/S and MAXATASE® by International Bio-Synthetics, Inc. Other proteases include Protease A; Protease B and proteases made by Genencor International, Inc., according to U.S. Pat. Nos. 5,204,015 and 5,244,791.

Amylases include, for example, alpha-amylases like RAPIDASE®, International BioSynthetics, Inc. and TERMAMYL®, Novo Industries.

Suitable lipase enzymes for detergent usage include those produced by microorganisms of the *Pseudomonas* group, such as *Pseudomonas stutzeri* ATCC 19154. This lipase is available from Amano Pharmaceutical Co. Ltd., under the trade name Lipase P "Amano". Other commercial lipases include Amano-CES, lipases ex *Chromobacter viscosum*, e.g. *Chromobacter viscosum* var. *lipolyticum* NRRLB 3673,

commercially available from Toyo Jozo Co., and further *Chromobacter viscosum* lipases from U.S. Biochemical Corp. and Disoynt Co., and lipases ex *Pseudomonas gladioli*. The LIPOLASE® enzyme derived from *Humicola lanuginosa* (commercially available from Novo Industries A/S) is a preferred lipase for use herein.

Peroxidase enzymes are used in combination with oxygen sources, e.g., percarbonate, perborate, persulfate, hydrogen peroxide, etc. They are used for "solution bleaching," i.e. to prevent transfer of dyes or pigments removed from substrates during wash operations to other substrates in the wash solution. Peroxidase enzymes are known in the art, and include, for example, horseradish peroxidase, ligninase, and haloperoxidase such as chloro- and bromo-peroxidase. Peroxidase-containing detergent compositions are disclosed, for example, in WO 1989 099813 A1.

Enzyme Stabilizers. The enzymes employed herein are stabilized by the presence of water-soluble sources of calcium and/or magnesium ions in the finished detergent compositions which provide such ions to the enzymes. (Calcium ions are generally somewhat more effective than magnesium ions and are preferred herein if only one type of cation is being used.) Additional stability can be provided by the presence of various other art-disclosed stabilizers, especially borate species, see U.S. Pat. No. 4,537,706, incorporated herein in its entirety. Typical detergents, especially liquids, will comprise from about 1 to about 30, preferably from about 2 to about 20, more preferably from about 5 to about 15, and most preferably from about 8 to about 12, millimoles of calcium ion per liter of finished composition. In solid detergent compositions the formulation can include a sufficient quantity of a water-soluble calcium ion source to provide such amounts in the laundry liquor. In the alternative, natural water hardness can suffice.

It is to be understood that the foregoing levels of calcium and/or magnesium ions are sufficient to provide enzyme stability. More calcium and/or magnesium ions can be added to the compositions to provide an additional measure of grease removal performance. Accordingly, as a general proposition the compositions herein will typically comprise from about 0.05% to about 2% by weight of a water-soluble source of calcium or magnesium ions, or both. The amount can vary, of course, with the amount and type of enzyme employed in the composition.

The compositions herein can also optionally, but preferably, contain various additional stabilizers, especially borate-type stabilizers. Typically, such stabilizers will be used at levels in the compositions from about 0.25% to about 10%, preferably from about 0.5% to about 5%, more preferably from about 0.75% to about 3%, by weight of boric acid or other borate compound capable of forming boric acid in the composition (calculated on the basis of boric acid). Boric acid is preferred, although other compounds such as boric oxide, borax and other alkali metal borates (e.g., sodium ortho-, meta- and pyroborate, and sodium pentaborate) are suitable. Substituted boric acids (e.g., phenylboronic acid, butane boronic acid, and p-bromo phenylboronic acid) can also be used in place of boric acid.

Builders

Polycarboxylic acids. Useful organic cobuilders are, for example, the polycarboxylic acids usable in the form of their sodium salts of polycarboxylic acids, wherein polycarboxylic acids are understood to be carboxylic acids that carry more than one acid function. These include, for example, citric acid, adipic acid, succinic acid, glutaric acid, malic acid, tartaric acid, maleic acid, fumaric acid, sugar acids, aminocarboxylic acids, nitrilotriacetic acid (NTA) and its

derivatives and mixtures thereof. Preferred salts are the salts of polycarboxylic acids such as citric acid, adipic acid, succinic acid, glutaric acid, tartaric acid, sugar acids and mixtures thereof.

Organic acids. Acids per se can also be used. Besides their building effect, the acids also typically have the property of an acidifying component and, hence also serve to establish a relatively low and mild pH in detergents or cleansing compositions. Citric acid, succinic acid, glutaric acid, adipic acid, gluconic acid and any mixtures thereof are particularly mentioned in this regard. Further suitable acidifiers are the known pH regulators such as sodium hydrogen carbonate and sodium hydrogen sulfate.

Polymers. Particularly suitable polymeric cobuilders are polyacrylates, which preferably have a molecular weight of 2,000 to 20,000 g/mol. By virtue of their superior solubility, preferred representatives of this group are again the short-chain polyacrylates, which have molecular weights of 2,000 to 10,000 g/mol and, more particularly, 3,000 to 5,000 g/mol. Suitable polymers can also include substances that consist partially or totally of vinyl alcohol units or its derivatives.

Further suitable copolymeric polycarboxylates are particularly those of acrylic acid with methacrylic acid and of acrylic acid or methacrylic acid with maleic acid. Copolymers of acrylic acid with maleic acid, which comprise 50 to 90 wt. % acrylic acid and 50 to 10 wt. % maleic acid, have proven to be particularly suitable. Their relative molecular weight, based on free acids, generally ranges from 2,000 to 70,000 g/mol, preferably 20,000 to 50,000 g/mol and especially 30,000 to 40,000 g/mol. The (co)polymeric polycarboxylates can be added either as an aqueous solution or preferably as powder. In order to improve the water solubility, the polymers can also comprise allylsulfonic acids as monomers, such as, for example, allyloxybenzene sulfonic acid and methallyl sulfonic acid as in the EP 0727448 B1.

Biodegradable polymers comprising more than two different monomer units are particularly preferred, examples being those comprising, as monomers, salts of acrylic acid and of maleic acid, and also vinyl alcohol or vinyl alcohol derivatives, as in DE 4300772 A1, or those comprising, as monomers, salts of acrylic acid and of 2-alkylallyl sulfonic acid, and also sugar derivatives. Further preferred copolymers are those that are described in German Patent Applications DE 4303320 A1 and DE 4417734 A1 and preferably include acrolein and acrylic acid/acrylic acid salts or acrolein and vinyl acetate as monomers.

Similarly, other preferred builders are polymeric aminodicarboxylic acids, salts or precursors thereof. Those polyaspartic acids or their salts and derivatives disclosed in German Patent Application DE 19540086 A1 as having a bleach-stabilizing action in addition to cobuilder properties are particularly preferred.

Further suitable builders are polyacetals that can be obtained by treating dialdehydes with polyol carboxylic acids that possess 5 to 7 carbon atoms and at least 3 hydroxyl groups, as described in European Patent Application EP 0280223 A1. Preferred polyacetals are obtained from dialdehydes like glyoxal, glutaraldehyde, terephthalaldehyde as well as their mixtures and from polycarboxylic acids like gluconic acid and/or glucoheptonic acid.

Carbohydrates. Further suitable organic cobuilders are dextrans, for example, oligomers or polymers of carbohydrates that can be obtained by the partial hydrolysis of starches. The hydrolysis can be carried out using typical processes, for example, acidic or enzymatic catalyzed processes. The hydrolysis products preferably have average

molecular weights in the range of 400 to 500,000 g/mol. A polysaccharide with a dextrose equivalent (DE) of 0.5 to 40 and, more particularly, 2 to 30 is preferred, the DE being an accepted measure of the reducing effect of a polysaccharide in comparison with dextrose, which has a DE of 100. Both maltodextrins with a DE between 3 and 20 and dry glucose syrups with a DE between 20 and 37 and also so-called yellow dextrans and white dextrans with relatively high molecular weights of 2,000 to 30,000 g/mol may be used. A preferred dextrin is described in British Patent Application 94 19 091.

The oxidized derivatives of such dextrans concern their reaction products with oxidizing compositions that are capable of oxidizing at least one alcohol function of the saccharide ring to the carboxylic acid function. Such oxidized dextrans and processes for their manufacture are known for example, from European Patent Applications EP 0232202 A1. A product oxidized at C6 of the saccharide ring can be particularly advantageous.

Oxydisuccinates and other derivatives of disuccinates, preferably ethylenediamine disuccinate are also further suitable cobuilders. Here, ethylene diamine-N,N'-disuccinate (EDDS), the synthesis of which is described for example, in U.S. Pat. No. 3,158,615, is preferably used in the form of its sodium or magnesium salts. In this context, glycerine disuccinates and glycerine trisuccinates are also particularly preferred, such as those described in U.S. Pat. No. 4,524, 009. Suitable addition quantities in zeolite-containing and/or silicate-containing formulations range from 3 to 15% by weight.

(Lactones. Other useful organic co-builders are, for example, acetylated hydroxycarboxylic acids and salts thereof which optionally may also be present in lactone form and which contain at least 4 carbon atoms, at least one hydroxyl group and at most two acid groups. Such cobuilders are described, for example, in International Patent Application WO 1995 020029 A1.

Bleaching Compounds, Bleaching Agents and Bleach Activators

The detergent compositions herein can optionally contain bleaching agents or bleaching compositions containing a bleaching agent and one or more bleach activators. When present, bleaching agents will typically be at levels of from about 1% to about 30%, more typically from about 5% to about 20%, of the detergent composition, especially for fabric laundering. If present, the amount of bleach activators will typically be from about 0.1% to about 60%, more typically from about 0.5% to about 40% of the bleaching composition comprising the bleaching agent-plus-bleach activator.

The bleaching agents used herein can be any of the bleaching agents useful for detergent compositions in textile cleaning, hard surface cleaning, or other cleaning purposes that are now known or become known. These include oxygen bleaches as well as other bleaching agents. Perborate bleaches, e.g., sodium perborate (e.g., mono- or tetra-hydrate) can be used herein.

Another category of bleaching agent that can be used without restriction encompasses percarboxylic acid bleaching agents and salts thereof. Suitable examples of this class of agents include magnesium monoperoxyphthalate hexahydrate, the magnesium salt of meta-chloro perbenzoic acid, 4-nonylamino-4-oxoperoxybutyric acid and diperoxydodecanedioic acid.

Peroxygen bleaching agents can also be used. Suitable peroxygen bleaching compounds include sodium carbonate peroxyhydrate and equivalent "percarbonate" bleaches,

sodium pyrophosphate peroxyhydrate, urea peroxyhydrate, and sodium peroxide. Persulfate bleach (e.g., OXONE®, manufactured commercially by DuPont) can also be used.

A preferred percarbonate bleach comprises dry particles having an average particle size in the range from about 500 micrometers to about 1,000 micrometers, not more than about 10% by weight of said particles being smaller than about 200 micrometers and not more than about 10% by weight of said particles being larger than about 1,250 micrometers. Optionally, the percarbonate can be coated with silicate, borate or water-soluble surfactants. Percarbonate is available from various commercial sources.

Mixtures of bleaching agents can also be used.

Peroxygen bleaching agents, the perborates, the percarbonates, etc., are preferably combined with bleach activators, which lead to the in situ production in aqueous solution (i.e., during the washing process) of the peroxy acid corresponding to the bleach activator. The nonanoyloxybenzene sulfonate (NOBS) and tetraacetyl ethylene diamine (TAED) activators are typical, and mixtures thereof can also be used.

Preferred amido-derived bleach activators include (6-octanamido-caproyl)oxybenzene-sulfonate, (6-nonanamido-caproyl)oxybenzenesulfonate, (6-decanamidocaproyl)oxybenzenesulfonate, and mixtures thereof.

Another class of bleach activators comprises the benzoxazin-type activators disclosed in U.S. Pat. No. 4,966,723, incorporated herein by reference.

Highly preferred lactam activators include benzoyl caprolactam, octanoyl caprolactam, 3,5,5-trimethylhexanoyl caprolactam, nonanoyl caprolactam, decanoyl caprolactam, undecenoyl caprolactam, benzoyl valerolactam, octanoyl valerolactam, decanoyl valerolactam, undecenoyl valerolactam, nonanoyl valerolactam, 3,5,5-trimethylhexanoyl valerolactam and mixtures thereof, optionally adsorbed into solid carriers, e.g. acyl caprolactams, preferably benzoyl caprolactam, adsorbed into sodium perborate.

Bleaching agents other than oxygen bleaching agents are also known in the art and can be utilized herein. One type of non-oxygen bleaching agent of particular interest includes photoactivated bleaching agents such as the sulfonated zinc and/or aluminum phthalocyanines. If used, detergent compositions will typically contain from about 0.025% to about 1.25%, by weight, of such bleaches, especially sulfonate zinc phthalocyanine.

If desired, the bleaching compounds can be catalyzed by means of a manganese compound. Such manganese-based catalysts are well known in the art and include $Mn^{IV}_2(u-O)_3(1,4,7\text{-trimethyl-1,4,7-triazacyclononane})_2(PF_6)_2$, $Mn^{III}_2(u-O)_1(u-OAc)_2(1,4,7\text{-trimethyl-1,4,7-triazacyclononane})_2(ClO_4)_2$, $Mn^{IV}_4(u-O)_6(1,4,7\text{-triazacyclononane})_4(ClO_4)_4$, $Mn^{III}Mn^{IV}_4(u-O)_1(u-OAc)_2(1,4,7\text{-trimethyl-1,4,7-triazacyclononane})_2(ClO_4)_3$, $Mn^{IV}_2(1,4,7\text{-trimethyl-1,4,7-triazacyclononane})-(OCH_3)_3(PF_6)$, and mixtures thereof.

As a practical matter, and not by way of limitation, the compositions and processes herein can be adjusted to provide on the order of at least one part per ten million of the active bleach catalyst species in the aqueous washing liquor, and will preferably provide from about 0.1 ppm to about 700 ppm, more preferably from about 1 ppm to about 500 ppm, of the catalyst species in the laundry liquor.

Polymeric Soil Release Agents

Any polymeric soil release agent known to those skilled in the art can optionally be employed in the detergent compositions and processes of this invention. Polymeric soil release agents are characterized by having both hydrophilic segments, to hydrophilize the surface of hydrophobic fibers, such as polyester and nylon, and hydrophobic segments, to

deposit upon hydrophobic fibers and remain adhered thereto through completion of washing and rinsing cycles and, thus, serve as an anchor for the hydrophilic segments. This can enable stains occurring subsequent to treatment with the soil release agent to be more easily cleaned in later washing procedures.

The polymeric soil release agents useful herein especially include those soil release agents having: (a) one or more nonionic hydrophile components consisting essentially of (i) polyoxyethylene segments with a degree of polymerization of at least 2, or (ii) oxypropylene or polyoxypropylene segments with a degree of polymerization of from 2 to 10, wherein said hydrophile segment does not encompass any oxypropylene unit unless it is bonded to adjacent moieties at each end by ether linkages, or (iii) a mixture of oxyalkylene units comprising oxyethylene and from 1 to about 30 oxypropylene units wherein said mixture contains a sufficient amount of oxyethylene units such that the hydrophile component has hydrophilicity great enough to increase the hydrophilicity of conventional polyester synthetic fiber surfaces upon deposit of the soil release agent on such surface, said hydrophile segments preferably comprising at least about 25% oxyethylene units and more preferably, especially for such components having about 20 to 30 oxypropylene units, at least about 50% oxyethylene units; or (b) one or more hydrophobe components comprising (i) C_3 oxyalkylene terephthalate segments, wherein, if said hydrophobe components also comprise oxyethylene terephthalate, the ratio of oxyethylene terephthalate: C_3 oxyalkylene terephthalate units is about 2:1 or lower, (ii) C_4 - C_6 alkylene or oxy C_4 - C_6 alkylene segments, or mixtures therein, (iii) poly(vinyl ester) segments, preferably polyvinyl acetate, having a degree of polymerization of at least 2, or (iv) C_1 - C_4 alkyl ether or C_4 hydroxyalkyl ether substituents, or mixtures therein, wherein said substituents are present in the form of C_1 - C_4 alkyl ether or C_4 hydroxyalkyl ether cellulose derivatives, or mixtures therein, and such cellulose derivatives are amphiphilic, whereby they have a sufficient level of C_1 - C_4 alkyl ether and/or C_4 hydroxyalkyl ether units to deposit upon conventional polyester synthetic fiber surfaces and retain a sufficient level of hydroxyls, once adhered to such conventional synthetic fiber surface, to increase fiber surface hydrophilicity, or a combination of (a) and (b).

Typically, the polyoxyethylene segments of (a) (i) will have a degree of polymerization of from about 200, although higher levels can be used, preferably from 3 to about 150, more preferably from 6 to about 100. Suitable oxy C_4 - C_6 alkylene hydrophobe segments include, but are not limited to, end-caps of polymeric soil release agents.

Polymeric soil release agents useful in the present invention also include cellulosic derivatives such as hydroxyether cellulosic polymers, copolymeric blocks of ethylene terephthalate or propylene terephthalate with polyethylene oxide or polypropylene oxide terephthalate, and the like. Such agents are commercially available and include hydroxyethers of cellulose such as METHOCEL® (Dow). Cellulosic soil release agents for use herein also include those selected from the group consisting of C_1 - C_4 alkyl and C_4 hydroxyalkyl cellulose.

Soil release agents characterized by poly(vinyl ester) hydrophobe segments include graft copolymers of poly(vinyl ester), e.g., C_1 - C_6 vinyl esters, preferably poly(vinyl acetate) grafted onto polyalkylene oxide backbones, such as polyethylene oxide backbones, see EP 0 219 048, incorporated herein in its entirety. Commercially available soil release agents of this kind include the SOKALAN® type of material, e.g., SOKALAN® HP-22, available from BASF.

One type of preferred soil release agent is a copolymer having random blocks of ethylene terephthalate and polyethylene oxide (PEO) terephthalate. The molecular weight of this polymeric soil release agent preferably is in the range of from about 25,000 to about 55,000.

Another preferred polymeric soil release agent is a polyester with repeat units of ethylene terephthalate units contains 10-15% by weight of ethylene terephthalate units together with 90-80% by weight of polyoxyethylene terephthalate units, derived from a polyoxyethylene glycol of average molecular weight 300-5,000. Examples of this polymer include the commercially available material ZELCON® 5126 (from DuPont) and MILEASE® T (from ICI).

Another preferred polymeric soil release agent is a sulfonated product of a substantially linear ester oligomer comprised of an oligomeric ester backbone of terephthaloyl and oxyalkyleneoxy repeat units and terminal moieties covalently attached to the backbone. These soil release agents are described fully in U.S. Pat. No. 4,968,451. Other suitable polymeric soil release agents include the terephthalate polyesters of U.S. Pat. No. 4,711,730, the anionic end-capped oligomeric esters of U.S. Pat. No. 4,721,580, the block polyester oligomeric compounds of U.S. Pat. No. 4,702,857, and anionic, especially sulfoaroyl, end-capped terephthalate esters of U.S. Pat. No. 4,877,896 all cited patents incorporated herein in their entirety.

Still another preferred soil release agent is an oligomer with repeat units of terephthaloyl units, sulfoisoterephthaloyl units, oxyethyleneoxy and oxy-1,2-propylene units. The repeat units form the backbone of the oligomer and are preferably terminated with modified isethionate end-caps. A particularly preferred soil release agent of this type comprises about one sulfoisophthaloyl unit, 5 terephthaloyl units, oxyethyleneoxy and oxy-1,2-propyleneoxy units in a ratio of from about 1.7 to about 1.8, and two end-cap units of sodium 2-(2-hydroxyethoxy)-ethanesulfonate. Said soil release agent also comprises from about 0.5% to about 20%, by weight of the oligomer, of a crystalline-reducing stabilizer, preferably selected from the group consisting of xylene sulfonate, cumene sulfonate, toluene sulfonate, and mixtures thereof.

If utilized, soil release agents will generally comprise from about 0.01% to about 10.0%, by weight, of the detergent compositions herein, typically from about 0.1% to about 5%, preferably from about 0.2% to about 3.0%.

Polymeric Dispersing Agents

Polymeric dispersing agents can advantageously be utilized at levels from about 0.1% to about 7%, by weight, in the detergent compositions herein, especially in the presence of zeolite and/or layered silicate builders. Suitable polymeric dispersing agents include polymeric polycarboxylates and polyethylene glycols, although others known in the art can also be used. It is believed, though it is not intended to be limited by theory, that polymeric dispersing agents enhance overall detergent builder performance, when used in combination with other builders (including lower molecular weight polycarboxylates) by crystal growth inhibition, particulate soil release peptization, and anti-redeposition.

Polymeric polycarboxylate materials can be prepared by polymerizing or copolymerizing suitable unsaturated monomers, preferably in their acid form. Unsaturated monomeric acids that can be polymerized to form suitable polymeric polycarboxylates include acrylic acid, maleic acid (or maleic anhydride), fumaric acid, itaconic acid, aconitic acid, mesaconic acid, citraconic acid and methylenemalononic acid. The presence in the polymeric polycarboxylates herein or monomeric segments, containing no carboxylate radicals such as

vinylmethyl ether, styrene, ethylene, etc. is suitable provided that such segments do not constitute more than about 40% by weight.

Particularly suitable polymeric polycarboxylates can be derived from acrylic acid. Such acrylic acid-based polymers which are useful herein are the water-soluble salts of polymerized acrylic acid. The average molecular weight of such polymers in the acid form preferably ranges from about 2,000 to 10,000, more preferably from about 4,000 to 7,000 and most preferably from about 4,000 to 5,000. Water-soluble salts of such acrylic acid polymers can include, for example, the alkali metal, ammonium and substituted ammonium salts. Soluble polymers of this type are known materials. Use of polyacrylates of this type in detergent compositions has been disclosed, for example U.S. Pat. No. 3,308,067.

Acrylic/maleic-based copolymers can also be used as a preferred component of the dispersing/anti-redeposition agent. Such materials include the water-soluble salts of copolymers of acrylic acid and maleic acid. The average molecular weight of such copolymers in the acid form preferably ranges from about 2,000 to 100,000, more preferably from about 5,000 to 75,000, most preferably from about 7,000 to 65,000. The ratio of acrylate to maleate segments in such copolymers will generally range from about 30:1 to about 1:1, more preferably from about 10:1 to 2:1. Water-soluble salts of such acrylic acid/maleic acid copolymers can include, for example, the alkali metal, ammonium and substituted ammonium salts. Soluble acrylate/maleate copolymers of this type are known materials which are described in EP 0193360 A1, which also describes such polymers comprising hydroxypropylacrylate. Still other useful dispersing agents include the maleic/acrylic/vinyl alcohol terpolymers, for example, a 45/45/10 terpolymer of acrylic/maleic/vinyl alcohol.

Another polymeric material which can be included is polyethylene glycol (PEG). PEG can exhibit dispersing agent performance as well as act as a clay soil removal-antiredeposition agent. Typical molecular weight ranges for these purposes range from about 500 to about 100,000, preferably from about 1,000 to about 50,000, more preferably from about 1,500 to about 10,000.

Polyaspartate and polyglutamate dispersing agents can also be used, especially in conjunction with zeolite builders. Dispersing agents such as polyaspartate preferably have a molecular weight (avg.) of about 10,000.

Foam Inhibitors/Sud Suppressors

Especially when used in automatic washing processes, it can be advantageous to add conventional foam inhibitors to the compositions. Suitable foam inhibitors include for example, soaps of natural or synthetic origin, which have a high content of C₁₈-C₂₄ fatty acids. Suitable non-surface-active types of foam inhibitors are, for example, organopolysiloxanes and mixtures thereof with microfine, optionally silanised silica and also paraffins, waxes, microcrystalline waxes and mixtures thereof with silanised silica or bis-stearyl ethylenediamide. Mixtures of various foam inhibitors, for example, mixtures of silicones, paraffins or waxes, are also used with advantage. Preferably, the foam inhibitors, especially silicone-containing and/or paraffin-containing foam inhibitors, are loaded onto a granular, water-soluble or dispersible carrier material. Especially in this case, mixtures of paraffins and bis-stearylethylene diamides are preferred.

Compounds for reducing or suppressing the formation of suds can be incorporated into the detergent compositions of the present invention. Suds suppression can be of particular

importance in the so-called "high concentration cleaning process" and in frontloading European-style washing machines.

A wide variety of materials can be used as suds suppressors, and suds suppressors are well known to those skilled in the art. See, for example, Kirk Othmer Encyclopedia of Chemical Technology, Third Edition, Volume 7, pages 430-447 (John Wiley & Sons, Inc., 1979). One category of suds suppressor of particular interest encompasses monocarboxylic fatty acid and soluble salts therein. The monocarboxylic fatty acids and salts thereof used as suds suppressor typically have hydrocarbyl chains of 10 to about 24 carbon atoms, preferably 12 to 18 carbon atoms. Suitable salts include the alkali metal salts such as sodium, potassium, and lithium salts, and ammonium and alkanolammonium salts.

The detergent compositions herein can also contain non-surfactant suds suppressors. These include, for example: high molecular weight hydrocarbons such as paraffin, fatty acid esters (e.g., fatty acid triglycerides), fatty acid esters of monovalent alcohols, aliphatic C_{18} - C_{40} ketones (e.g., stearone), etc. Other suds inhibitors include N-alkylated amino triazines such as tri- to hexa-alkylmelamines or di- to tetra-alkyldiamine chlortriazines formed as products of cyanuric chloride with two or three moles of a primary or secondary amine containing 1 to 24 carbon atoms, propylene oxide, and monostearyl phosphates such as monostearyl alcohol phosphate ester and monostearyl di-alkali metal (e.g., K, Na, and Li) phosphates and phosphate esters. The hydrocarbons such as paraffin and haloparaffin can be utilized in liquid form. The liquid hydrocarbons will be liquid at room temperature and atmospheric pressure, and will have a pour point in the range of about -40° C. and about 50° C., and a minimum boiling point not less than about 110° C. (atmospheric pressure). It is also known to utilize waxy hydrocarbons, preferably having a melting point below about 100° C. Hydrocarbon suds suppressors are known in the art and include aliphatic, alicyclic, aromatic, and heterocyclic saturated or unsaturated hydrocarbons having from about 12 to about 70 carbon atoms. The term "paraffin," as used in this suds suppressor discussion, is intended to include mixtures of true paraffins and cyclic hydrocarbons.

Another preferred category of non-surfactant suds suppressors comprises silicone suds suppressors. This category includes the use of polyorganosiloxane oils, such as polydimethylsiloxane, dispersions or emulsions of polyorganosiloxane oils or resins, and combinations of polyorganosiloxane with silica particles wherein the polyorganosiloxane is chemisorbed or fused onto the silica. Silicone suds suppressors are well known in the art.

Other silicone suds suppressors are disclosed in U.S. Pat. No. 3,455,839, incorporated herein in its entirety, which relates to compositions and processes for defoaming aqueous solutions by incorporating therein small amounts of polydimethylsiloxane fluids.

Mixtures of silicone and silanated silica are described, for instance, in DE-OS 2124526, incorporated herein in its entirety. Silicone defoamers and suds controlling agents in granular detergent compositions are disclosed in U.S. Pat. No. 4,652,392, incorporated herein in its entirety.

In the preferred silicone suds suppressor used herein, the solvent for a continuous phase is made up of certain polyethylene glycols or polyethylene-polypropylene glycol copolymers or mixtures thereof (preferred), or polypropylene glycol. The primary silicone suds suppressor is branched/crosslinked and preferably not linear.

The silicone suds suppressor herein preferably comprises polyethylene glycol and a copolymer of polyethylene glycol/

polypropylene glycol, all having an average molecular weight of less than about 1,000, preferably between about 100 and 800. The polyethylene glycol and polyethylene/polypropylene copolymers herein have a solubility in water at room temperature of more than about 2 weight %, preferably more than about 5 weight %.

The preferred solvent herein is polyethylene glycol having an average molecular weight of less than about 1,000, more preferably between about 100 and 800, most preferably between 200 and 400, and a copolymer of polyethylene glycol/polypropylene glycol, preferably PPG 200/PEG 300. Preferred is a weight ratio of between about 1:1 and 1:10, most preferably between 1:3 and 1:6, of polyethylene glycol:copolymer of polyethylenepolypropylene glycol.

The preferred silicone suds suppressors used herein do not contain polypropylene glycol, particularly of 4,000 molecular weight. They also preferably do not contain block copolymers of ethylene oxide and propylene oxide, like PLURONIC® L101.

Other suds suppressors useful herein comprise the secondary alcohols (e.g., 2-alkyl alkanols) and mixtures of such alcohols with silicone oils. The secondary alcohols include the C_6 - C_{16} alkyl alcohols having a C_1 - C_{16} chain. A preferred alcohol is 2-butyl octanol, which is available from Condea under the trademark ISOFOL® 12. Mixtures of secondary alcohols are available under the trademark ISALCHEM® 123 from Enichem. Mixed suds suppressors typically comprise mixtures of alcohol+silicone at a weight ratio of 1:5 to 5:1.

The compositions herein will generally comprise from 0% to about 5% of suds suppressor. When utilized as suds suppressors, monocarboxylic fatty acids, and salts therein, will be present typically in amounts up to about 5%, by weight, of the detergent composition. Preferably, from about 0.5% to about 3% of fatty monocarboxylate suds suppressor is utilized. Silicone suds suppressors are typically utilized in amounts up to about 2.0%, by weight, of the detergent composition, although higher amounts can be used. This upper limit is practical in nature, due primarily to concern with keeping costs minimized and effectiveness of lower amounts for effectively controlling sudsing. Preferably from about 0.01% to about 1% of silicone suds suppressor is used, more preferably from about 0.25% to about 0.5%. As used herein, these weight percentage values include any silica that can be utilized in combination with polyorganosiloxane, as well as any adjunct materials that can be utilized. Monostearyl phosphate suds suppressors are generally utilized in amounts ranging from about 0.1% to about 2%, by weight, of the composition. Hydrocarbon suds suppressors are typically utilized in amounts ranging from about 0.01% to about 5.0%, although higher levels can be used. The alcohol suds suppressors are typically used at 0.2%-3% by weight of the finished compositions.

Sequestrants and Chelating Agents

The salts of polyphosphonic acid can be considered as sequestrants or as stabilizers, particularly for peroxy compounds and enzymes, which are sensitive towards heavy metal ions. Here, the sodium salts of, for example, 1-hydroxyethane-1,1-diphosphonate, diethylenetriamine pentamethylene phosphonate or ethylenediamine tetramethylene phosphonate are used in amounts of 0.1 to 5 wt. %.

The detergent compositions herein can also optionally contain one or more iron and/or manganese chelating agents. Such chelating agents can be selected from the group consisting of amino carboxylates, amino phosphonates, polyfunctionally-substituted aromatic chelating agents and mixtures therein, all as hereinafter defined. Without intend-

ing to be bound by theory, it is believed that the benefit of these materials is due in part to their exceptional ability to remove iron and manganese ions from washing solutions by formation of soluble chelates. It is understood that some of the detergent builders described hereinbefore can function as chelating agents and if such detergent builder is present in a sufficient quantity, it can provide both functions.

Amino carboxylates useful as optional chelating agents include ethylenediaminetetracetates, N-hydroxyethylethylenediaminetriacetates, nitrilotriacetates, ethylenediamine tetrapropionates, triethylenetetraaminehexacetates, diethylenetriaminepentaacetates, and ethanoldiglycines, alkali metal, ammonium, and substituted ammonium salts therein and mixtures therein.

Amino phosphonates are also suitable for use as chelating agents in the compositions of the invention when at least low levels of total phosphorus are permitted in detergent compositions, and include ethylenediaminetetrakis (methylene phosphonates) as DEQUEST. Preferred, these amino phosphonates to not contain alkyl or alkenyl groups with more than about 6 carbon atoms.

Polyfunctionally-substituted aromatic chelating agents are also useful in the compositions herein. Preferred compounds of this type in acid form are dihydroxydisulfobenzenes such as 1,2-dihydroxy-3,5-disulfobenzene.

A preferred biodegradable chelator for use herein is ethylenediamine disuccinate ("EDDS"), especially the [S,S] isomer.

If utilized, these chelating agents will generally comprise from about 0.1% to about 10% by weight of the detergent compositions herein. More preferably, if utilized, the chelating agents will comprise from about 0.1% to about 3.0% by weight of such compositions.

Clay Soil Removal/Anti-Redeposition Agents

The detergent compositions of the present invention can also optionally contain water-soluble ethoxylated amines having clay soil removal and antiredeposition properties. Granular detergent compositions which contain these compounds typically contain from about 0.01% to about 10.0% by weight of the water-soluble ethoxylates amines; liquid detergent compositions typically contain about 0.01% to about 5%.

The most preferred soil release and anti-redeposition agent is ethoxylated tetraethylenepentamine. Exemplary ethoxylated amines are further described in U.S. Pat. No. 4,597,898. Other groups of preferred clay soil removal-antiredeposition agents are the cationic compounds disclosed in EP 0111965 A1, the ethoxylated amine polymers disclosed in EP 0111984 A1, the zwitterionic polymers disclosed in EP 0112592 A1, and the amine oxides disclosed in U.S. Pat. No. 4,548,744. Another type of preferred antiredeposition agent includes the carboxy methyl cellulose (CMC) materials. These materials are well known in the art. Graying Inhibitors

Graying inhibitors have the function of maintaining the dirt that was removed from the fibers suspended in the washing liquor, thereby preventing the dirt from resettling. Water-soluble colloids of mostly organic nature are suitable for this, for example, the water-soluble salts of (co)polymeric carboxylic acids, glue, gelatins, salts of ether carboxylic acids or ether sulfonic acids of starches or celluloses, or salts of acidic sulfuric acid esters of celluloses or starches. Water-soluble, acid group-containing polyamides are also suitable for this purpose. Moreover, soluble starch preparations and others can be used as the above-mentioned starch products, e.g., degraded starches, aldehyde starches etc. Polyvinyl pyrrolidone can also be used. Preference, how-

ever, is given to the use of cellulose ethers such as carboxymethyl cellulose (Na salt), methyl cellulose, hydroxyalkyl celluloses and mixed ethers such as methyl hydroxyethyl cellulose, methyl hydroxypropyl cellulose, methyl carboxymethyl cellulose and mixtures thereof, as well as polyvinyl pyrrolidone, which can be added, for example, in amounts of 0.1 to 5 wt. %, based on the composition.

Optical Brighteners and UV Adsorbers

Any optical brighteners or other brightening or whitening agents known in the art can be incorporated at levels typically from about 0.05% to about 1.2%, by weight, into the detergent compositions herein. Commercial optical brighteners which can be useful in the present invention can be classified into subgroups, which include, but are not necessarily limited to, derivatives of stilbene, pyrazoline, coumarin, carboxylic acid, methinecyanines, dibenzothiphenene-5,5-dioxide, azoles, 5- and 6-membered-ring heterocycles, and other miscellaneous agents.

Preferred brighteners include the PHORWHITE® series of brighteners from Verona. Other brighteners disclosed in this reference include: Tinopal® UNPA, Tinopal CBS and Tinopal 5BM; available from Ciba-Geigy; Artic White® CC and Artic White CWD, available from Hilton-Davis; the 2-(4-stryl-phenyl)-2H-naphthol [1,2-d]triazoles; 4,4'-bis-(1,2,3-triazol-2-yl)-stilbenes; 4,4'-bis(stryl)bisphenyls; and the aminocoumarins. Specific examples of these brighteners include 4-methyl-7-diethyl-amino coumarin; 1,2-bis(-venzimidazol-2-yl)ethylene; 1,3-diphenyl-phrazolines; 2,5-bis(benzoxazol-2-yl)thiophene; 2-stryl-naph-[1,2-d] oxazole; and 2-(stilbene-4-yl)-2H-naphtho-[1,2-d]triazole. Anionic brighteners are preferred herein.

The compositions may comprise e.g., derivatives of diaminostilbene disulfonic acid or alkali metal salts thereof as the optical brighteners. Suitable optical brighteners are, for example, salts of 4,4'-bis-(2-anilino-4-morpholino-1,3,5-triazinyl-6-amino)stilbene-2,2'-disulfonic acid or compounds of similar structure which contain a diethanolamino group, a methylamino group, an anilino group or a 2-methoxyethylamino group instead of the morpholino group. Brighteners of the substituted diphenylstyryl type may also be present, for example, the alkali metal salts of 4,4'-bis(2-sulfostyryl)diphenyl, 4,4'-bis(4-chloro-3-sulfostyryl)diphenyl or 4-(4-chlorostyryl)-4'-(2-sulfostyryl)diphenyl. Mixtures of the mentioned brighteners may also be used.

In addition, UV absorbers may also be added. These are compounds with distinct absorption abilities for ultra violet radiation, which contribute as UV stabilizers as well as to improve the light stability of colorants and pigments both for textile fibers as well as for the skin of the wearer of textile products by protecting against the UV radiation that penetrates the fabric. In general, the efficient radiationless deactivating compounds are derivatives of benzophenone, substituted with hydroxyl and/or alkoxy groups, mostly in position(s) 2 and/or 4. Also suitable are substituted benzotriazoles, additionally acrylates that are phenyl-substituted in position 3 (cinnamic acid derivatives), optionally with cyano groups in position 2, salicylates, organic Ni complexes, as well as natural substances such as umbelliferone and the endogenous urocanic acid. In a preferred embodiment, the UV absorbers absorb UV-A and UV-B radiation as well as possible UV-C radiation and re-emit light with blue wavelengths, such that they additionally have an optical brightening effect. Preferred UV absorbers encompass triazine derivatives, e.g., hydroxyaryl-1,3,5-triazine, sulfonated 1,3,5-triazine, o-hydroxyphenylbenzotriazole

and 2-aryl-2H-benzotriazole as well as bis(anilino)stilbene disulfonic acid and their derivatives. Ultra violet absorbing pigments like titanium dioxide can also be used as UV absorbers.

Dye Transfer Inhibiting Agents

The detergent compositions of the present invention can also include one or more materials effective for inhibiting the transfer of dyes from one fabric to another during the cleaning process. Generally, such dye transfer inhibiting agents include polyvinyl pyrrolidone polymers, polyamine N-oxide polymers, copolymers of N-vinylpyrrolidone and N-vinylimidazole, manganese phthalocyanine, peroxidases, and mixtures thereof. If used, these agents typically comprise from about 0.01% to about 10% by weight of the composition, preferably from about 0.01% to about 5%, and more preferably from about 0.05% to about 2%.

More specifically, the polyamine N-oxide polymers preferred for use herein are described in U.S. Pat. No. 6,491,728, incorporated herein by reference.

Any polymer backbone can be used as long as the amine oxide polymer formed is water-soluble and has dye transfer inhibiting properties. Examples of suitable polymeric backbones are polyvinyls, polyalkylenes, polyesters, polyethers, polyamide, polyimides, polyacrylates and mixtures thereof. These polymers include random or block copolymers where one monomer type is an amine N-oxide and the other monomer type is an N-oxide. The amine N-oxide polymers typically have a ratio of amine to the amine N-oxide of 10:1 to 1:1,000,000. However, the number of amine oxide groups present in the polyamine oxide polymer can be varied by appropriate copolymerization or by an appropriate degree of N-oxidation. The polyamine oxides can be obtained in almost any degree of polymerization. Typically, the average molecular weight is within the range of 500 to 1,000,000; more preferred 1,000 to 500,000; most preferred 5,000 to 100,000. This preferred class of materials can be referred to as "PVNO".

The most preferred polyamine N-oxide useful in the detergent compositions herein is poly(4-vinylpyridine-N-oxide) which has an average molecular weight of about 50,000 and an amine to amine N-oxide ratio of about 1:4.

Copolymers of N-vinylpyrrolidone and N-vinylimidazole polymers (referred to as a class as "PVPVI") are also preferred for use herein. Preferably the PVPVI has an average molecular weight range from 5,000 to 1,000,000, more preferably from 5,000 to 200,000, and most preferably from 10,000 to 20,000. The PVPVI copolymers typically have a molar ratio of N-vinylimidazole to N-vinylpyrrolidone from 1:1 to 0.2:1, more preferably from 0.8:1 to 0.3:1, most preferably from 0.6:1 to 0.4:1. These copolymers can be either linear or branched.

The present invention compositions also can employ a polyvinylpyrrolidone ("PVP") having an average molecular weight of from about 5,000 to about 400,000, preferably from about 5,000 to about 200,000, and more preferably from about 5,000 to about 50,000. PVP's are known to persons skilled in the detergent field. Compositions containing PVP can also contain polyethylene glycol ("PEG") having an average molecular weight from about 500 to about 100,000, preferably from about 1,000 to about 10,000. Preferably, the ratio of PEG to PVP on a ppm basis delivered in wash solutions is from about 2:1 to about 50:1, and more preferably from about 3:1 to about 10:1.

The detergent compositions herein can also optionally contain from about 0.005% to 5% by weight of certain types of hydrophilic optical brighteners which also provide a dye

transfer inhibition action. If used, the compositions herein will preferably comprise from about 0.01% to 1% by weight of such optical brighteners.

One preferred brightener is 4,4'-bis[(4-anilino-6-(N-2-bis-hydroxyethyl)-s-triazine-2-yl)amino]-2,2'-stilbenedisulfonic acid and disodium salt. This particular brightener species is commercially marketed under the trade name Tinopal-UNPA-GX® by Ciba-Geigy Corporation. Tinopal-UNPA-GX is the preferred hydrophilic optical brightener useful in the detergent compositions herein.

Another preferred brightener is 4,4'-bis[(4-anilino-6-(N-2-hydroxyethyl-N-methylamino)-s-triazine-2-yl)amino]-2,2'-stilbenedisulfonic acid disodium salt. This particular brightener species is commercially marketed under the trade name Tinopal 5BM-GX® by Ciba-Geigy Corporation.

Another preferred brightener is 4,4'-bis[(4-anilino-6-morpholino-s-triazine-2-yl)amino]-2,2'-stilbenedisulfonic acid, sodium salt. This particular brightener species is commercially marketed under the trade name Tinopal AMS-GX® by Ciba Geigy Corporation.

The specific optical brightener species selected for use in the present invention provide especially effective dye transfer inhibition performance benefits when used in combination with the selected polymeric dye transfer inhibiting agents hereinbefore described. The combination of such selected polymeric materials (e.g., PVNO and/or PVPVI) with such selected optical brighteners (e.g., Tinopal UNPA-GX, Tinopal 5BM-GX and/or Tinopal AMS-GX) provides significantly better dye transfer inhibition in aqueous wash solutions than does either of these two detergent composition components when used alone. Without being bound by theory, it is believed that such brighteners work this way because they have high affinity for fabrics in the wash solution and therefore deposit relatively quickly on these fabrics. The extent to which brighteners deposit on fabrics in the wash solution can be defined by a parameter called the "exhaustion coefficient". The exhaustion coefficient is in general as the ratio of a) the brightener material deposited on fabric to b) the initial brightener concentration in the wash liquor. Brighteners with relatively high exhaustion coefficients are the most suitable for inhibiting dye transfer in the context of the present invention.

Of course, it will be appreciated that other, conventional optical brightener types of compounds can optionally be used in the present compositions to provide conventional fabric "brightness" benefits, rather than a true dye transfer inhibiting effect. Such usage is conventional and well-known to detergent formulations.

Thickeners

The compositions can also comprise common thickeners and anti-deposition compositions as well as viscosity regulators such as polyacrylates, polycarboxylic acids, polysaccharides and their derivatives, polyurethanes, polyvinyl pyrrolidones, castor oil derivatives, polyamine derivatives such as quaternized and/or ethoxylated hexamethylenediamines as well as any mixtures thereof. Preferred compositions have a viscosity below 10,000 mPa*s, measured with a Brookfield viscosimeter at a temperature of 20° C. and a shear rate of 50 min⁻¹.

Inorganic Salts

Further suitable ingredients of the composition are water-soluble inorganic salts such as bicarbonates, carbonates, amorphous silicates or mixtures of these; alkali carbonate and amorphous silicate are particularly used, principally sodium silicate with a molar ratio Na₂O:SiO₂ of 1:1 to 1:4.5, preferably of 1:2 to 1:3.5. Preferred compositions comprise alkaline salts, builders and/or cobuilders, preferably sodium

carbonate, zeolite, crystalline, layered sodium silicates and/or trisodium citrate, in amounts of 0.5 to 70 wt. %, preferably 0.5 to 50 wt. %, particularly 0.5 to 30 wt. % anhydrous substance.

Perfumes and Colorants

The compositions can comprise further typical detergent and cleansing composition ingredients such as perfumes and/or colorants, wherein such colorants are preferred that leave no or negligible coloration on the fabrics being washed. Preferred amounts of the totality of the added colorants are below 1 wt. %, preferably below 0.1 wt. %, based on the composition. The compositions can also comprise white pigments such as e.g., TiO_2 .

INDUSTRIAL APPLICATION

Another object of the present invention refers to a method for manufacturing a liquid detergent composition comprising the following steps:

- (i) providing the components for manufacturing the composition, optionally in aqueous solution;
- (ii) providing the dispersion of microcapsules according to Claim 1 or obtained according to the process of Claim 2;
- (iii) blending the components and the dispersion to obtain the final composition.

Finally, the invention also covers the use of microcapsules for manufacturing liquid detergent compositions.

EXAMPLES

Example 1

A microcapsule of this invention was prepared using guanidine carbonate as the multi-functional nucleophile and CapsulTM starch as the capsule formation aid.

More specifically, 196 g of a fragrance, W-Cap (Symrise, Teterboro, NJ) was weighed out in a 250 ml beaker and combined with 10.4 g of hexamethylene diisocyanate (Desmodur N-3400, Covestro Corporation) to form an oil phase. In a separate 800 ml beaker, a solution (375 g) containing 2% of CapsulTM starch (commercially available from Ingredion Inc.) in water to form the aqueous phase. The oil phase was then emulsified into the aqueous phase to form the fragrance emulsion under shearing (Ultra Turrax, T-50, commercially available from IKA Werke) at 3500 rpm between 20 and 60 seconds.

The fragrance emulsion was placed with an overhead mixer and agitated at 600 rpm while 25 g of a 15% guanidine carbonate solution added incrementally. The capsule slurry was cured by heat at 70° C. for at least 3 hours.

The capsule slurry of this example was disbursed in commercially available unfragranced liquid laundry detergent at a dose of 0.3% w/w until homogeneous. The liquid laundry detergent contains no structuring agents. Samples were aged in an oven for 7 days at 40 C to simulate long term stability. This is shown in FIG. 2.

Example 2

A microcapsule of this invention was prepared using guanidine carbonate as the multi-functional nucleophile and CapsulTM starch as the capsule formation aid.

More specifically, 98 g of a fragrance, Tomcap (Symrise, Teterboro, NJ) was weighed out in a 250 ml beaker and combined with 98 g of vegetable oil triglycerides and 10.4 grams of a mixture of hexamethylene diisocyanate and

diphenylmethane diisocyanate (Desmodur N-3400 and Mondur M, both from Covestro Corporation) to form an oil phase. In a separate 800 ml beaker, a solution (375 g) containing 2% of CapsulTM starch (commercially available from Ingredion Inc.) in water to form the aqueous phase. The oil phase was then emulsified into the aqueous phase to form the fragrance emulsion under shearing (Ultra Turrax, T-50, commercially available from IKA Werke) at 3500 rpm between 30 to 60 seconds. The particle size was measured on a Mastersizer[®]3000 (Malvern Instruments|117 Flanders Road, Westborough, MA) particle size analyzer to produce desired median size of 5 to 50 microns.

The fragrance emulsion was placed with an overhead mixer and agitated at 600 rpm while 25 g of a 15% guanidine carbonate solution added incrementally. The capsule slurry was cured by heat at 70° C. for 3 hours.

The capsule slurry of this example was disbursed in commercially available unfragranced liquid laundry detergent at a dose of 0.3% w/w until homogeneous. The liquid laundry detergent contains no structuring agents. Samples were aged in an oven for 7 days at 40 C to simulate long term stability. This is shown in FIG. 3.

Example 3

A microcapsule of this invention was prepared using guanidine carbonate as the multi-functional nucleophile and Hi-CapTM starch as the capsule formation aid.

More specifically, 98 g of a fragrance, Red Berry (Symrise, Teterboro, NJ) was weighed out in a 250 ml beaker and combined with 98 g of vegetable oil triglycerides and 10.4 grams of a mixture of hexamethylene diisocyanate and diphenylmethane diisocyanate (Desmodur N-3400 and Mondur M, both from Covestro Corporation) to form an oil phase. In a separate 800 ml beaker, a solution (375 g) containing 2% of Hi-CapTM100 starch (commercially available from Ingredion Inc.) in water to form the aqueous phase. The oil phase was then emulsified into the aqueous phase to form the fragrance emulsion under shearing (Ultra Turrax, T-50, commercially available from IKA Werke) at 3500 rpm between 30 to 60 seconds. The particle size was measured on a Mastersizer[®]3000 (Malvern Instruments|117 Flanders Road, Westborough, MA) particle size analyzer to produce desired median size of 5 to 50 microns.

The fragrance emulsion was placed with an overhead mixer and agitated at 600 rpm while 25 g of a 15% guanidine carbonate solution added incrementally. The capsule slurry was cured by heat at 70° C. for 3 hours.

The capsule slurry of this example was disbursed in commercially available unfragranced liquid laundry detergent at a dose of 0.3% w/w until homogeneous. The liquid laundry detergent contains no structuring agents. Samples were aged in an oven for 7 days at 40 C to simulate long term stability. This is shown in FIG. 4.

Example 4

A microcapsule of this invention was prepared using guanidine carbonate as the multi-functional nucleophile and CapsulTM starch as the capsule formation aid. A cationic quaternary was added during final process step.

More specifically, 98 g of a fragrance, Red Berry (Symrise, Teterboro, NJ) was weighed out in a 250 ml beaker and combined with 98 grams of vegetable oil triglycerides and 10.4 grams of a mixture of hexamethylene diisocyanate and diphenylmethane diisocyanate (Desmodur N-3400 and Mondur M, both from Covestro Corporation) to form an oil

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phase. In a separate 800 ml beaker, a solution (375 g) containing 2% of HI-Cap™100 starch (commercially available from Ingredion Inc.) in water to form the aqueous phase. The oil phase was then emulsified into the aqueous phase to form the fragrance emulsion under shearing (Ultra Turrax, T-50, commercially available from IKA Werke) at 3500 rpm between 30 to 60 seconds. The particle size was measured on a Mastersizer®3000 (Malvern Instruments|117 Flanders Road, Westborough, MA) particle size analyzer to produce desired median size of 5 to 50 microns.

The fragrance emulsion was placed with an overhead mixer and agitated at 600 rpm while 25 grams of a 15% Guanidine carbonate solution added incrementally. The capsule slurry was cured by heat at 70° C. for 3 hours. A cationic polymer (Salcare SC-60, BASF) was added to the capsule slurry as a solid powder or as a aqueous solution during final cure.

BRIEF DESCRIPTION OF THE FIGURES

FIG. 1

Symcap G Type (polyurea) microcapsules showing agglomeration in an unstructured liquid laundry base.

FIG. 2

Capsules prepared in Example 1 (right), disbursed in unstructured Liquid Laundry Detergent after 1 week at 40° C.

FIG. 3

Capsules prepared in Example 2 (right), disbursed in unstructured Liquid Laundry Detergent after 1 week at 40° C.

FIG. 4

Capsules prepared in Example 3 (right), disbursed in unstructured Liquid Laundry Detergent after 1 week at 40° C.

Consumer Product Formulations

TABLE 1

Cleaner, APC liquid, alkaline pH 8-10 (Amounts in % b.w.)	
Ingredients (INCI)	Amount
Aqua	59.06
Tri Sodium Citrate Dihydrate	3.00
Sodium Laureth Sulfate	30.00
Trideceth-9	5.00
Ethanol	2.00
Citric Acid 10% solution	0.24
1,2-pentanediol (Hydrolite ® 5)	0.50
Mixture of 5-Chloro-2-methyl-2H-isothiazol-3-one and 2-Methyl-2H-isothiazol-3-one	0.10
Microcapsules according to Example 4	0.30

TABLE 2

Fabric softener (Amounts in % b.w.)	
Ingredients (INCI)	Amount
Aqua	72.10
Dialkylester ammonium methosulfate	16.60
Polydimethylsiloxane	0.30
Magnesiumchloride	10.00
1,2-pentanediol (Hydrolite ® 5)	0.60
Mixture of 5-Chloro-2-methyl-2H-isothiazol-3-one and 2-Methyl-2H-isothiazol-3-one	0.10
Microcapsules according to Example 4	0.40

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TABLE 3

Liquid detergent (Amounts in % b.w.)	
Ingredients	Amount
Deionized water	39.60
Optical brightener	0.10
Coconut fatty acids (C12-C18)	7.50
Potassium hydroxide 50% solution	4.30
Propane-1,2-diol	5.00
Fatty alcohols C12-C15, 8 EO	12.00
Na-salt of secondary alkyl sulfonates (C13-C17)	17.00
Triethanolamine	2.00
Trisodium citrate dihydrate	5.00
Dequest 2066 Diethylenetriamine	3.00
penta(methylene phosphonic acid)	
Ethanol	3.00
Enzymes	0.70
1,2-pentanediol (Hydrolite ® 5)	0.50
Microcapsules according to Example 4	1.00

TABLE 4

Liquid detergent concentrate (Amounts in % b.w.)	
Ingredients	Amount
Deionized water	12.9
Coconut fatty acids (C12-C18)	10.0
Fatty alcohols C12-C15, 8 EO	26.0
Na-salt of secondary alkyl sulfonates (C13-C17)	26.5
Triethanol amine	8.5
Na-salt of fatty alcohol sulfates C12-C14	3.0
Ethanol	5.5
Urea	4.5
Enzymes	0.9
Citric acid	1.0
1,2-pentanediol (Hydrolite ® 5)	0.7
Microcapsules according to Example 4	0.8

TABLE 5

Toilet cleaner (Amounts in % b.w.)	
Ingredients	Amount
Water	93.0
Kelzan ASX-T	0.5
Parafin sulfonate, sodium salt	1.0
Citric acid	5.0
Colorant (FD & C Yellow No. 6)	0.1
1,2-pentanediol (Hydrolite ® 5)	0.3
Preservative (Benzisothiazolinone, Glutaral)	0.05
Microcapsules according to Example 4	0.6

TABLE 6

Dish washing concentrate (Amounts in % b.w.)	
Ingredients	Amount
Sodium laurylsulfate	31.0
Propane-1,2-diole	6.0
Ethyl alcohol 96%	7.0
Palm tree glucosides	6.0
Coco betaine	18.0
1,2-pentanediol (Hydrolite ® 5)	0.4
Microcapsules according to Example 4	0.5
Water	31.6

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TABLE 7

Dish washing concentrate (Amounts in % b.w.)	
Ingredients	Amount
Palm tree glucosides	4.0
Sodium lauryl sulfate	45.0
Coco betaine	8.0
Ethyl alcohol 96%	1.0
Colorant (C.I. Pigment Blue 15)	0.05
1,2-pentanediol (Hydrolite ® 5)	0.2
Microcapsules according to Example 4	0.7
Water	Ad 100

TABLE 8

Solution for wet wipes (Amounts in % b.w.)		
Ingredients	INCI	Amount
SymSol ® PF-3	Water (Aqua). Pentylene Glycol. Sodium Lauryl Sulfoacetate. SodiumOleoyl Sarcosinate. Sodium Chloride. Disodium Sulfoacetate. SodiumOleate. Sodium Sulfate	2.00
Dragosantol ® 100	Bisabolol	0.10
Glycerol 99.5 P.	Glycerol	5.00
Water	Water (Aqua)	Ad 100
Hydrolite ® 5	1,2-pentanediol	5.00
D-Panthenol 75 W	Panthenol	0.80
DragoCalm ®	Water (Aqua). Glycerol. Avena Sativa (Oat) Kernel Extract	1.00
Witch Hazel-Distillate	Hamamelis Virginiana (Witch Hazel) Water. Water (Aqua). Alcohol	1.00
Allplant Essence ® Org.	Pelargonium Graveolens	1.00
Rose Geranium P	Flower/Leaf/Stem Water	
Preservative	Phenoxyethanol	0.30
Microcapsules according to Example 4		0.50

What claimed is:

1. A process for making polyurea microcapsules, comprising the following steps:

- providing an oil phase comprising at least one aliphatic and/or aromatic di- and/or polyisocyanate and optionally at least one oil component;
- providing a first aqueous phase comprising at least one modified biopolymer colloidal protection agent;
- providing a second aqueous phase comprising guanidine carbonate and at least one cationic surface active agent;

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(d) blending the oil phase and the first aqueous phase to form an emulsion;

(e) adding the second aqueous phase to the emulsion formed in step (d) to form a dispersion of crude microcapsules;

(f) curing the mixture; and optionally

(g) removing the solvent(s) to obtain the microcapsules.

2. The process of claim 1, wherein said di- and/or polyisocyanate is selected from the group consisting of methylenediphenyl diisocyanate (MDI); toluol diisocyanate (TDI); hexamethylene diisocyanate (HDI); isophorone diisocyanate (IPDI); 4,4-dicyclohexylmethan diisocyanate (H12MDI); and mixtures thereof.

3. The process of claim 1, said oil phase further comprising one or more actives for encapsulation.

4. The process of claim 3, wherein said active is a fragrance or perfume oil.

5. The process of claim 1, wherein said colloidal protection agent is selected from the group consisting of chemically modified biopolymers, gum Arabic or cellulose, and mixtures thereof.

6. The process of claim 1, said first aqueous phase further comprising at least one non-interfering emulsifier.

7. The process of claim 6, wherein said at least one non-interfering emulsifier is selected from the group consisting of non-ionic, anionic, amphoteric, and cationic surfactants, and mixtures thereof.

8. The process of claim 1, wherein said emulsion is formed by subjecting the blend of said oil phase and said first aqueous phase to high-speed shearing.

9. The process of claim 1, wherein said emulsion is blended with said second aqueous phase and the mixture thus obtained is subjected to high-speed shearing.

10. The process of claim 1, said second aqueous phase further comprising at least one surface active agent.

11. The process of claim 1, wherein the curing takes place at elevated temperatures of from about 50 to about 90° C. over a period of from about 1 to about 12 hours.

12. The process of claim 5, wherein the chemically modified biopolymers are chemically modified starches.

13. The process of claim 1, wherein the solvent(s) are removed to obtain the microcapsules.

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